

2DE/2DE3: Differential Equations: Part B

Dr J. S. Keeler

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Chapter 0

Preliminaries

0.1. Lecture plan

The **approximate** plan for this (half) lecture-course is

1. Introduction (Lectures 1-2)
2. 1-dimensional dynamical systems
 - Stability of fixed-points and constructing the phase-portrait (Lecture 3-4)
 - 1-dimensional bifurcations (Lectures 5-6)
3. 2-dimensional linear dynamical systems
 - Linear systems and types of equilibria (Lectures 7-8)
4. 2-dimensional nonlinear dynamical systems
 - Nonlinear systems and constructing the phase portrait (Lectures 9-10)
 - Examples (Lectures 11-12)
 - Calculating stable/unstable manifolds (Lectures 13-14)
5. Periodic-orbits and Hamiltonian systems
 - Poincare-Bendixon theorem (Lectures 15-16)
 - Hamiltonian systems (Lectures 17-18)

0.2. Office hours

My office hours are **10am Mondays** and **15pm Thursdays**.

0.3. Reading list

- Nonlinear dynamics and Chaos by Steven Strogatz
- Nonlinear Systems by Phillip Drazin
- Elements of applied bifurcation theory by Yuri Kuzentsov

0.4. Assignments and Guided study sessions

There will be a weekly exercise sheet for you to attempt. The due-dates for the assignments for this part of the course are

- Assignment 1: Available 04/02/26, Due 11/02/26 (Part A)
- Assignment 2: Available 18/02/26, Due 25/02/26 (Part B)

- Assignment 3: Available 11/03/26, Due 25/03/26 (Part A)
- Assignment 4: Available 25/03/26, Due 29/04/26 (Part B)

They will be released at least a week before the due-date but I will aim to make it exactly a week before.

Chapter 1

Introduction

1.1. What is a dynamical system?

The notion of a dynamical system is the mathematical formalisation of the general scientific concept of a deterministic process. The future and past states of many physical, chemical, biological, economical and even social systems can be predicted to a certain extent by knowing their present state and the laws governing their evolution. Provided these laws do not change in time, the behaviour of such a system could be considered as completely defined by its initial state. Let's think about a couple of simple examples.

Example 1.1.1: Example 1: A first-order linear ODE

Find the particular solution of

$$\dot{x} = -x, \quad x(t=0) = x_0. \quad (1.1.1)$$

This can be solved via elementary techniques:

$$\begin{aligned} \Rightarrow \frac{dx}{dt} &= -x \\ \Rightarrow \frac{1}{x} dx &= -dt \\ \Rightarrow \int \frac{1}{x} dx &= - \int dt \\ \Rightarrow \log x &= -t + C \\ \Rightarrow x &= Ae^{-t} \\ \Rightarrow x(t=0) &= x_0 \in \mathbb{R} \\ \Rightarrow x &= x_0 e^{-t} \end{aligned}$$

From a dynamical systems perspective this tells us that **for all** $x_0 \in \mathbb{R}$, $x(t) \rightarrow 0$, i.e. the end state is independent of the initial conditions.

Example 1.1.2: Example 2: A second-order linear ODE

Find the particular solution of

$$\ddot{x} + 2\dot{x} + 2 = 0, \quad x(t=0) = x_0, \dot{x}(t=0) = y_0. \quad (1.1.2)$$

Again, this can be solved via elementary techniques:

$$\begin{aligned}
 \Rightarrow -1-x &\Rightarrow \ddot{x} + 2\dot{x} + 2 = 0 \\
 \Rightarrow x &= Ae^{\sigma t} \\
 \Rightarrow \sigma^2 + 2\sigma + 2 &= 0 \\
 \Rightarrow \sigma &= -1 \pm i \\
 \Rightarrow x &= e^{-t}(A_1 \cos(t) + A_2 \sin(t)) \\
 \Rightarrow x(t=0) = x_0 \in \mathbb{R}, \quad \dot{x}(t=0) = y_0 \in \mathbb{R} \\
 \Rightarrow x &= e^{-t}(x_0 \cos(t) + (x_0 + y_0) \sin(t))
 \end{aligned}$$

From a dynamical systems perspective this tells us that **for all** $(x_0, y_0) \in \mathbb{R}^2$, $x(t), \dot{x}(t) \rightarrow 0$, i.e. **the end state is independent of the initial conditions**.

Example 1.1.3: Example 3: A first-order nonlinear ODE

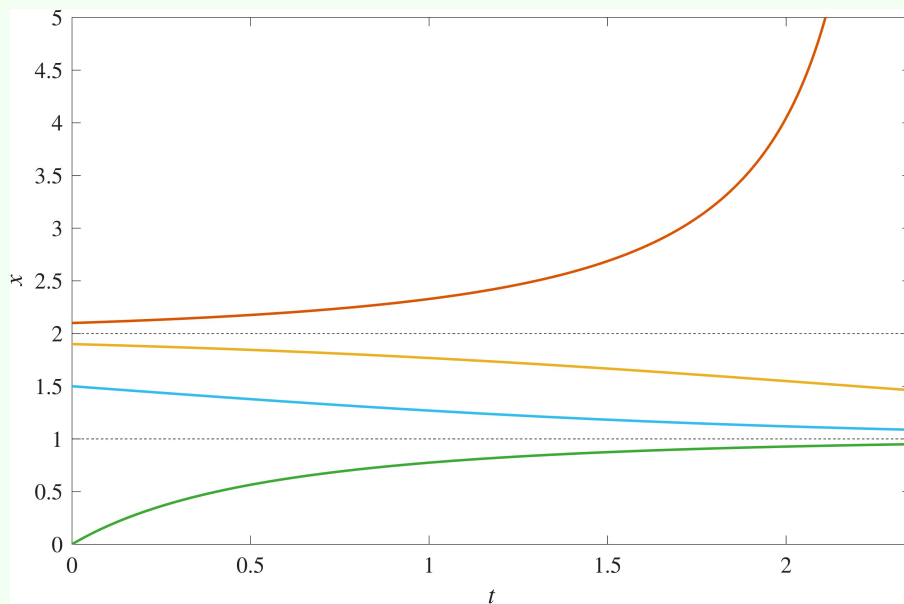
Find the particular solution of

$$\frac{dx}{dt} = -(1-x)(x-2), \quad x(t=0) = x_0. \quad (1.1.3)$$

Again, this can be solved via elementary techniques but we have to work a little harder:

$$\begin{aligned}
 \Rightarrow \dot{x} &= -(1-x)(x-2) \\
 \Rightarrow \frac{1}{(1-x)(x-2)} dx &= -dt \\
 \Rightarrow \log|x-1| - \log|x-2| &= -t + C \\
 \Rightarrow \log\left|\frac{x-1}{x-2}\right| &= -t + C \\
 \Rightarrow \frac{x-1}{x-2} &= Ae^{-t} \\
 \Rightarrow x &= \frac{1-2Ae^{-t}}{1-Ae^{-t}} \\
 \Rightarrow x &= \frac{1-2Ae^{-t}}{1-Ae^{-t}} \\
 \Rightarrow x &= \frac{(x_0-2) - 2(x_0-1)e^{-t}}{(x_0-2) - (x_0-1)e^{-t}}
 \end{aligned}$$

It is not as clear what happens to the system as $t \rightarrow \infty$. We can plot some individual solutions for different values of x_0 :



As these plots show, if $x_0 > 2$, then the solution blows up, otherwise it tends to 1. As the problem is nonlinear there are different possible outcomes depending on the initial condition.

Thus, the notion of a dynamical system comprises of

1. A set of possible states of the system: a **state space/phase space**
2. An **evolution law** that evolves the state of the system through ...
3. ...**time**

We will explore these concepts in more detail in this chapter. From a more fundamental level the overarching aims of dynamical systems theory are to answer the following questions:

- How does the end-state of the system as $t \rightarrow \infty$ depend on the initial conditions?
- Are there any states that do not change as t is varied? (invariant solutions)

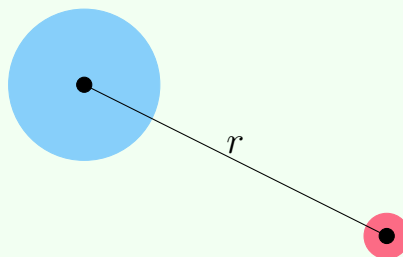
1.2. Examples

We will now highlight a number of different examples.

Example 1.2.1: Example 1: Two-body problem

This is the classical problem of understanding the motion of two objects that are subject to a central force. In the context of gravity this force is an inverse square law.

$$\ddot{\mathbf{r}} = - (GM/r^2) \mathbf{e}_r$$



This equation is highly nonlinear, but can be solved (with some beautiful transformations) so that it can be deduced that the following end outcomes are possible. The smaller object will form an orbit that takes the form of either a

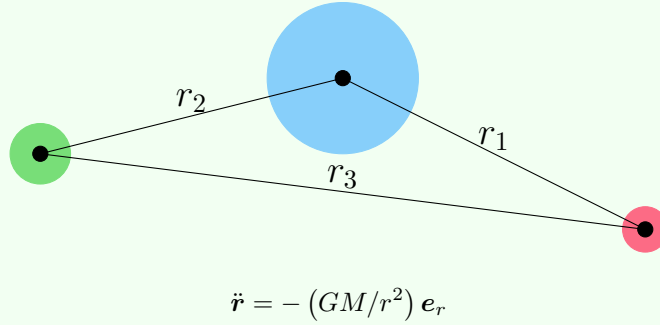
1. Circle
2. or Ellipse
3. or Parabola
4. or Hyperbola.

The dynamical systems question that we ask is how can we predict the outcome?

Example 1.2.2: Example 2: Three-body problem

$$\begin{aligned}\ddot{\mathbf{r}}_1 &= -Gm_2 \frac{(\mathbf{r}_1 - \mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3} - Gm_3 \frac{(\mathbf{r}_1 - \mathbf{r}_3)}{|\mathbf{r}_1 - \mathbf{r}_3|^3} \\ \ddot{\mathbf{r}}_2 &= -Gm_1 \frac{(\mathbf{r}_2 - \mathbf{r}_1)}{|\mathbf{r}_2 - \mathbf{r}_1|^3} - Gm_3 \frac{(\mathbf{r}_2 - \mathbf{r}_3)}{|\mathbf{r}_2 - \mathbf{r}_3|^3} \\ \ddot{\mathbf{r}}_3 &= -Gm_1 \frac{(\mathbf{r}_3 - \mathbf{r}_1)}{|\mathbf{r}_3 - \mathbf{r}_1|^3} - Gm_2 \frac{(\mathbf{r}_3 - \mathbf{r}_2)}{|\mathbf{r}_3 - \mathbf{r}_2|^3}\end{aligned}$$

Now we add another body to the two-body problem. The system is nonlinear and there are now no closed-form solutions. We use dynamical systems theory to understand what possible outcomes there are and whether they are stable.

**Example 1.2.3: Example 3: Is an explicit solution always worth it?**

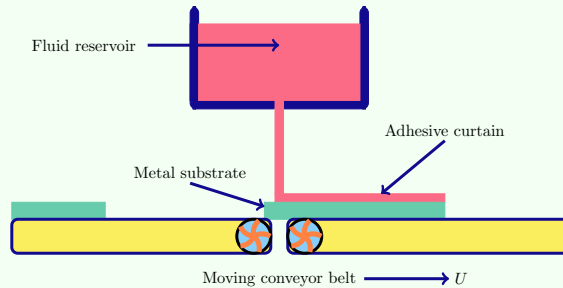
A one-dimensional nonlinear ODE of the form:

$$\dot{x} = (x+1)(x+2)(x+3).$$

This system has a closed-form analytic solution. However, the solution requires a non-trivial amount of work to obtain; can we instead glean information about such systems by just examining the fixed-points of the system, i.e. when $\dot{x} = 0$? This will form a large part of the lecture course.

Example 1.2.4: Example 3: Dip-coating

Dip-coating is an industrial process for coating materials. The material is placed on a conveyor belt and then painted with a viscous fluid. When the conveyor belt is too fast the coating is uneven and the material is wasted. From a dynamical systems perspective, we can model the system as an ODE for the angle the fluid makes to the air, θ , and the height, h . By analysing this system, what speeds can we choose so that the plate is spread evenly?



$$\begin{aligned}\frac{d^2\theta}{ds^2} &= 1 - \frac{Ca}{h^2} \\ \frac{dh}{ds} &= \cos(\theta),\end{aligned}$$

where Ca is a constant roughly proportional to the speed and s is the length taken along the fluid boundary.

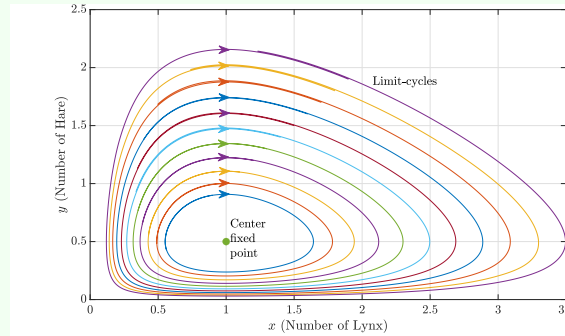
Example 1.2.5: Example 4: Lotka-Volterra system

The Lotka-Volterra system describes the dynamics of a population of prey and predators. Historically it was developed for modelling the dynamics of Lynx (predator) and Hare (prey) in Canada. The dynamical system can be described by the two-dimensional nonlinear set of ODE:

$$\dot{x} = \alpha x - \beta xy, \quad (1.2.1)$$

$$\dot{y} = -\gamma y + \delta xy, \quad (1.2.2)$$

where $\alpha, \beta, \gamma, \delta \in \mathbb{R}$ are fixed parameters. In this example there is a fixed point that is surrounded by **periodic-orbit** invariant solutions. The fixed point is an example of a **center**. The key aspect from a dynamical systems perspective is understanding what parameters and initial conditions support a sustainable population balance.



Example 1.2.6: Example 5: Lorentz system

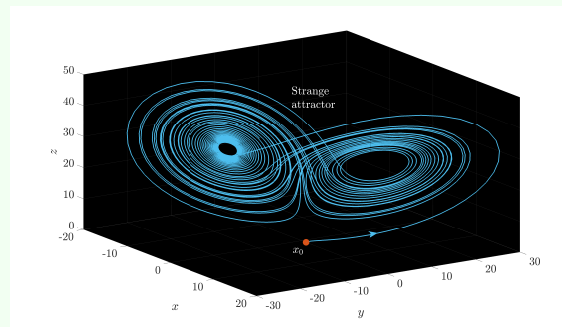
In 1963, Edward Lorenz formulated a simplified mathematical model to describe atmospheric convection. He was attempting to model the way air moves when heated from below and cooled from above. Without getting into the details his dynamical system can be described by the three-dimensional ODE

$$\dot{x} = \sigma(y - x), \quad (1.2.3)$$

$$\dot{y} = x(\rho - z) - y, \quad (1.2.4)$$

$$\dot{z} = xy - \beta z, \quad (1.2.5)$$

where $\sigma, \rho, \beta \in \mathbb{R}$ are fixed parameters. This was one of the first examples of **chaos**. Here the system has a very sensitive dependence on initial conditions and the limiting orbit is an example of a **strange attractor** invariant solution.

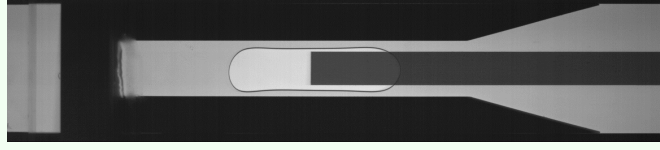


This was the first example studied of a dynamical system exhibiting chaos.

Example 1.2.7: Example 6: Air bubbles

In microfluidics, air bubbles occur naturally and it is important that we can control them. However, the bubble can change shape as it moves and it is not clear what shape and configuration the bubble will settle down to. The governing equations are now described by a PDE and the system

is infinitely-dimensional. Fortunately, ideas taken from simple two-dimensional systems, that we study here, can be implemented on this more difficult problem so we can understand the eventual outcome of the system



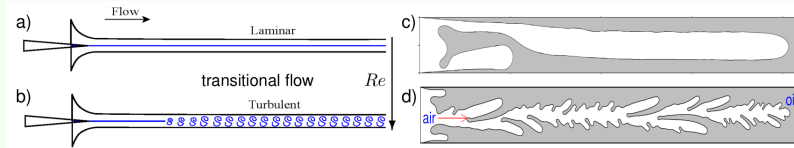
The system can be described, rather abstractly as

$$\dot{\mathbf{u}} = \text{Some nasty nonlinear function of } \mathbf{u}$$

where $\mathbf{u}$ represents the field of unknowns.

Example 1.2.8: Example 7: Turbulence

One of the most famous unsolved problems is that of when pipe-flow transitions from laminar to turbulent flow and what value of the Reynolds number (proportional to the speed) this occurs. This is also an infinite-dimensional system and state-of-the-art techniques for understanding this phenomena are continually being developed. Recent advances including using data science and machine learning by treating the problem as an infinite-dimensional dynamical-system and provide some promise. $\dot{\mathbf{u}} = \text{Some nasty nonlinear function of } \mathbf{u}$

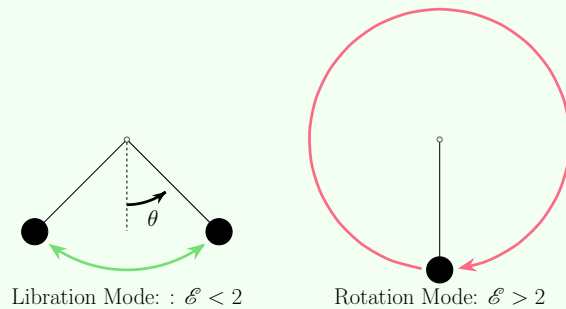


1.3. Definitions and notation

There is a distinct vocabulary in Dynamical systems theory that we use to describe system, solutions and qualitative behaviour. In this section we will introduce some of the key vocabulary through the discussion of a specific example: the nonlinear pendulum.

Example 1.3.1: Illustrative example: Nonlinear damped pendulum

We now shall examine the nonlinear damped-pendulum. Depending on how much ‘energy’ the pendulum has initially, there are two modes; a libration mode and a rotation mode:



The angle, θ , is modelled by the nonlinear ODE

$$\ddot{\theta} + \varepsilon \dot{\theta} + \sin(\theta) = 0, \quad (1.3.1)$$

or

$$\frac{d^2\theta}{dt^2} + \varepsilon \frac{d\theta}{dt} + \sin(\theta) = 0. \quad (1.3.2)$$

We also need to state the initial condition

$$\theta(t=0) = \theta_0, \dot{\theta}(t=0) = v. \quad (1.3.3)$$

From this incredibly basic physical system we can elucidate all of the main concepts/definitions that we will use for the rest of the course. Let's identify some important concepts/definitions.

Definition 1.3.1: Nonlinearity

First, the system is **nonlinear**:

$$\frac{d^2\theta}{dt^2} + \varepsilon \frac{d\theta}{dt} + \underbrace{\sin(\theta)}_{\text{Nonlinear}} = 0. \quad (1.3.4)$$

In general, a system is nonlinear if the dependent variable appears as a power or nonlinear function. Nonlinearity means a number of things:

- There is probably (almost certainly) **no closed-form analytic solution**
- There are potentially **multiple end states** of the system.

Definition 1.3.2: The dimensionality of a dynamical system

Next, the system is **second-order** as the highest derivative is second order

$$\underbrace{\frac{d^2\theta}{dt^2}}_{\text{Second-order}} + \varepsilon \frac{d\theta}{dt} + \sin(\theta) = 0. \quad (1.3.5)$$

This means that the dynamical system is **two-dimensional**; it is not enough to just have knowledge of θ in this system to understand it's current state; we also need $\dot{\theta}$. In fact we can write the **second-order** ODE as a **two-dimensional system** of ODE by writing

$$x = \theta, \quad y = \dot{\theta}, \quad (1.3.6)$$

so that the dynamical system becomes

$$\dot{x} = y, \quad (1.3.7)$$

$$\dot{y} = -\varepsilon y - \sin x. \quad (1.3.8)$$

- It is very useful to pose an n -dimensional ODE in the form

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

where $\mathbf{x}$ and $\mathbf{f}$ are n -dimensional vectors.

- It may sound obvious, but it is important to note that an n -dimensional system will require n initial conditions.

Definition 1.3.3: Autonomous

A dynamical system is **autonomous** if there is no explicit time dependence in the system. For example, if we added some time-periodic forcing to the pendulum (for example we could blow it) the system is

$$\frac{d^2\theta}{dt^2} + \varepsilon \frac{d\theta}{dt} + \sin(\theta) = \underbrace{\cos(t)}_{\text{Explicit time-dependence}}. \quad (1.3.9)$$

This system is **non-autonomous**; the laws of evolution change as t is varied. But, we can easily convert this to an autonomous system by adding another a phase variable and increasing the dimension of the system by 1, i.e.

$$\dot{x} = y, \quad (1.3.10)$$

$$\dot{y} = -\varepsilon y - \sin x + \cos(z), \quad (1.3.11)$$

$$\dot{z} = 1. \quad (1.3.12)$$

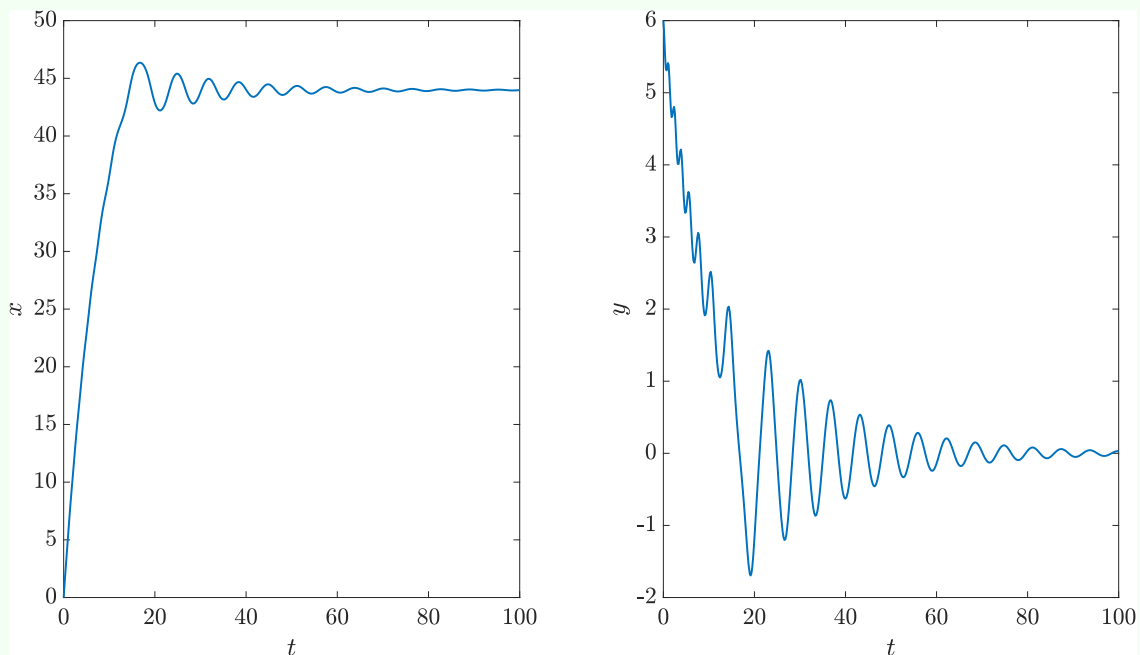
So for a given ODE we can now describe it as nonlinear/linear, the dimension of an ODE and whether it is autonomous/non-autonomous.

1.3.1 Visualisation of the solutions

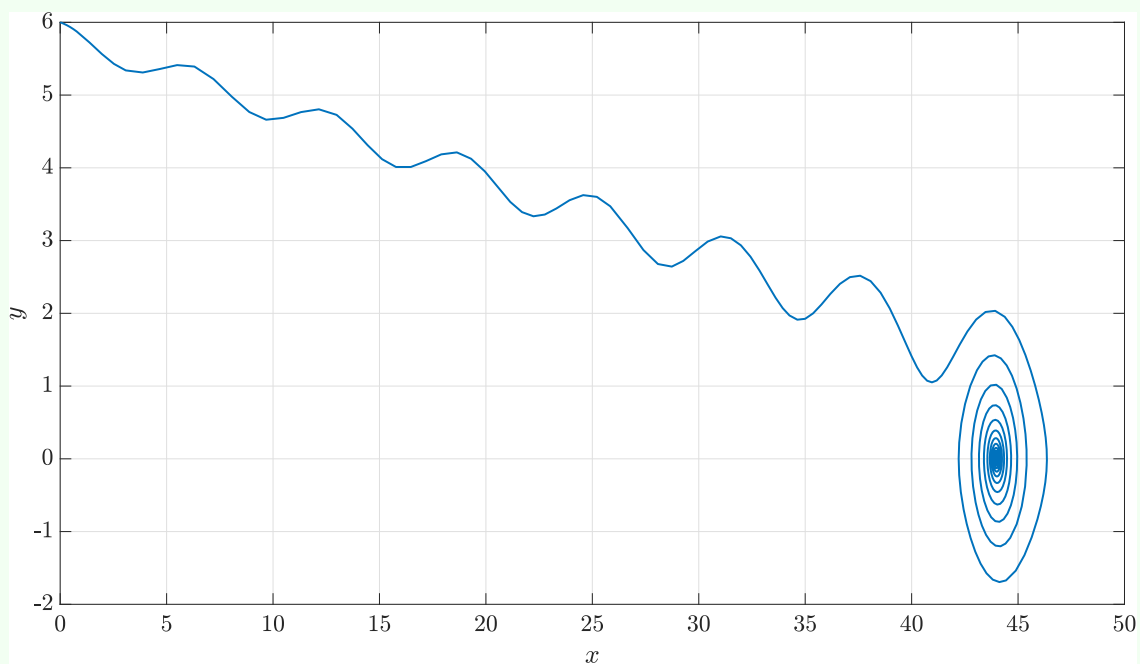
Before we go into detail as to how we solve these, it is useful to imagine yourself as an experimentalist, without any (or at least a little) mathematical knowledge. In this scenario the experimentalist wants to know what happens and the end state of the system without any regard to the beautiful mathematics involved. How can they view or categorise what happens? In this subsection we shall introduce some vocabulary that will help us describe solutions to the dynamical system.

Definition 1.3.4: Orbits, solution curves and phase space

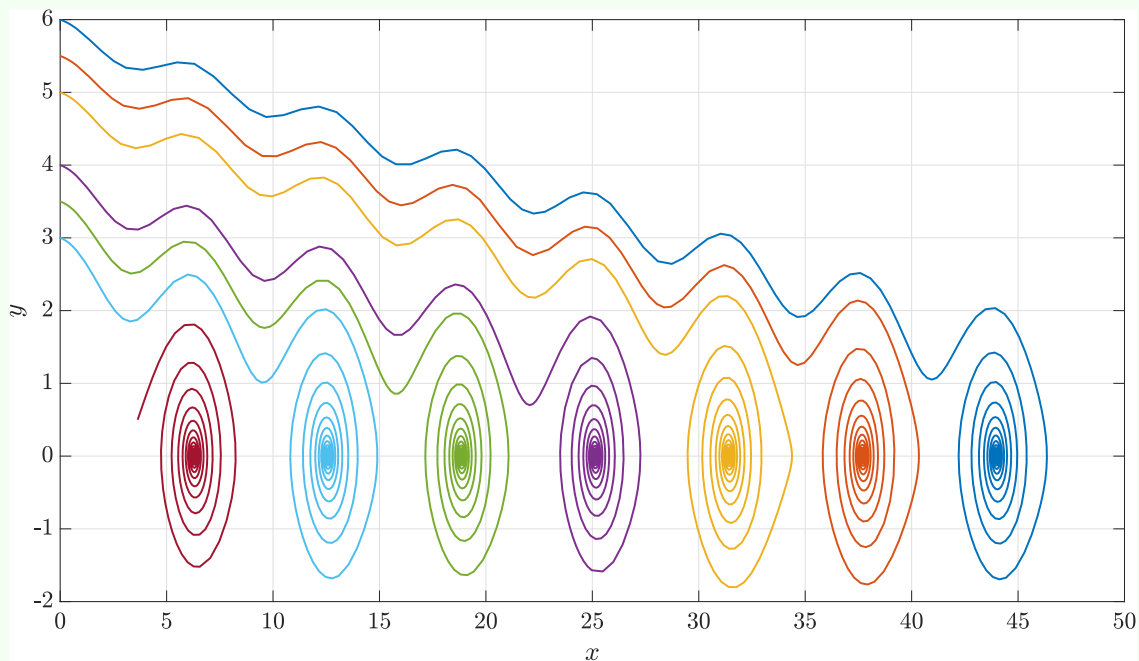
Let's pick an initial state of the pendulum by specifying an initial condition $(x(t=0), y(t=0))$ and then find $(x(t), y(t))$ by solving the system (numerically). We could plot these solutions as a function of t ; these are what we call **solution curves**.



Alternatively, we could plot these in **phase space** by plotting individual **orbits**. The phase space is two-dimensional and we plot each orbit in the (x, y) plane; each curve is parameterised by t .



The complete set of orbits is called the complete **phase portrait**; once we have this we know what happens to the system for **all initial conditions** and are job is done.



For any given dynamical system, we could complete the phase portrait by continuously changing the initial conditions and then computing orbits; in general this is **costly** and **labourious**. It turns out we do not need to do this and there is a cleverer way to obtain the universal phase portrait.

1.3.2 Invariant solutions and stability

A mathematician, on the other hand would prefer to make some predictions about the final outcomes based on the equations. Their key tool that a mathematician would use is to find **invariant solutions**.

By calculating and locating **invariant solutions** in phase space and then determining their stability allows us to sketch a phase portrait without having to plot hundreds of individual orbits.

We will define the concept of invariant solutions or sets in detail later on in the course, but broadly speaking these are solutions to the dynamical system that do not change in time. There are many types of invariant solutions; for example **periodic-cycles**, **invariant-loci** or even **strange attractors**.

Definition 1.3.5: Fixed points

The simplest type are **fixed points** which are found by setting the time-derivatives to zero and then solving the resulting algebraic set of equations. For the nonlinear pendulum this means solving

$$0 = \dot{y}, \quad (1.3.13)$$

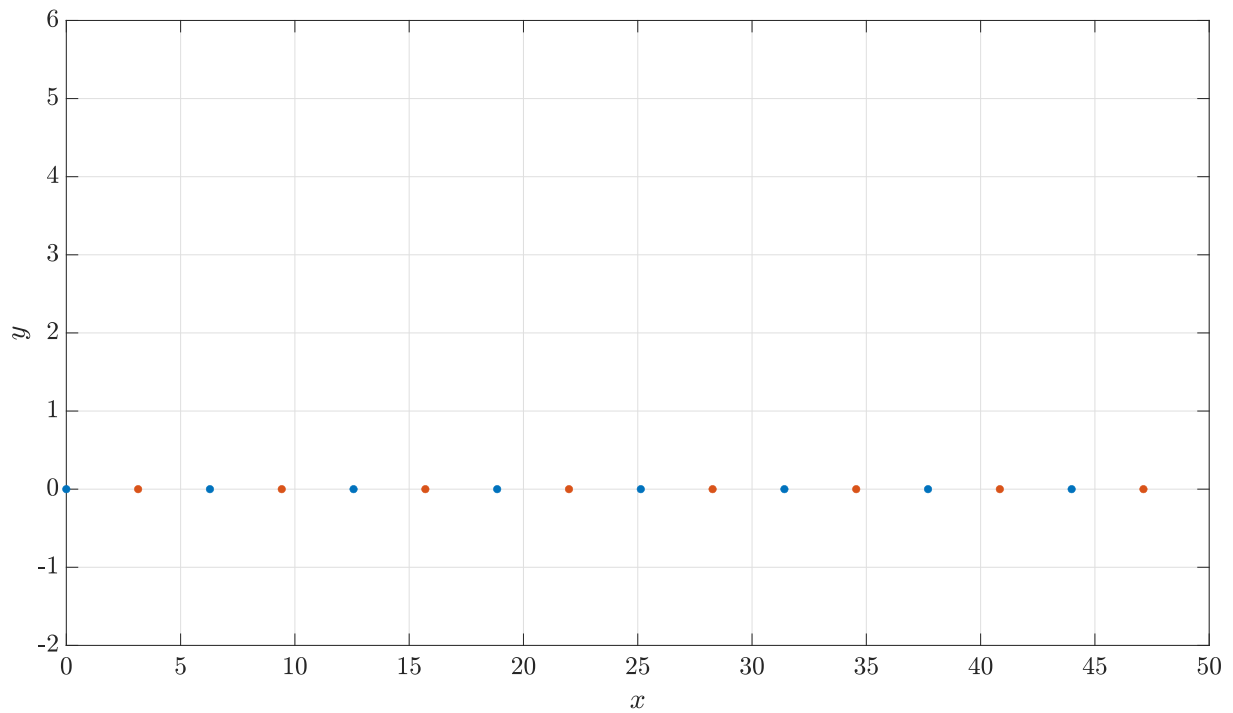
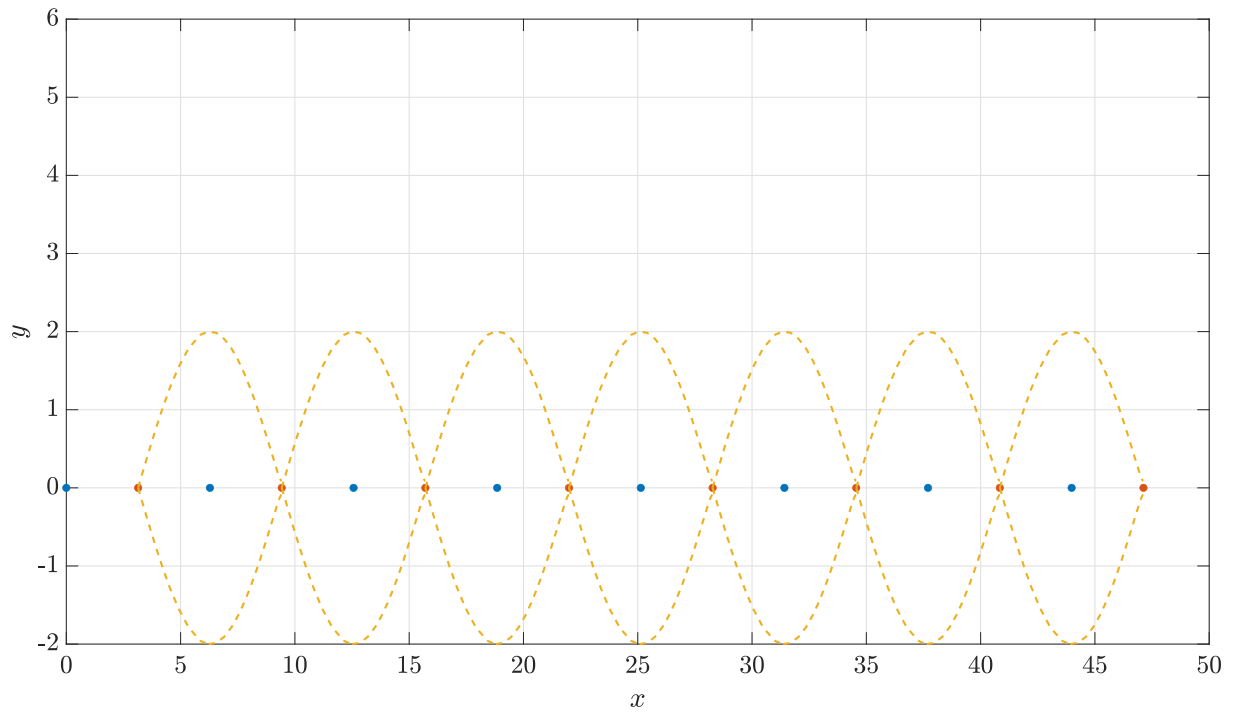
$$0 = -\varepsilon y - \sin x. \quad (1.3.14)$$

which has the infinite set of solutions:

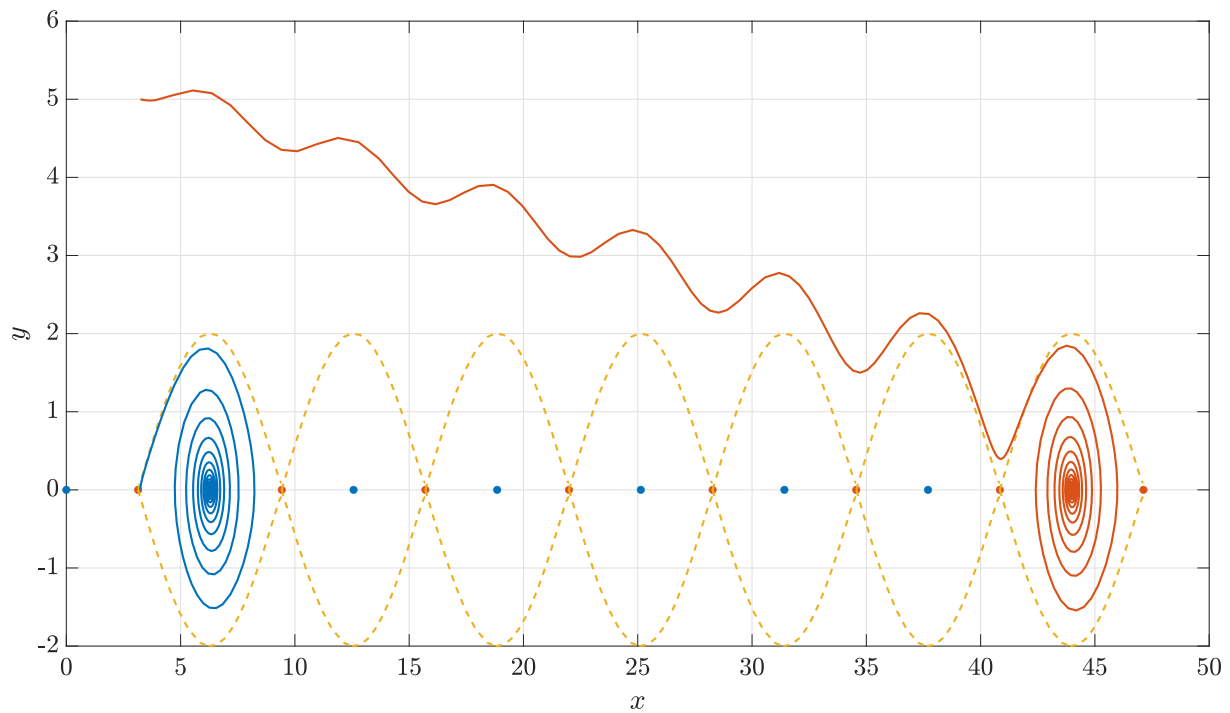
$$\mathbf{x}_s = (x_s, y_s) = (n\pi, 0), \quad n \in \mathbb{Z} \quad (1.3.15)$$

If n is even then the pendulum is at the downwards position; this fixed point is **stable** and is a long term dynamical outcome. If n is odd then the fixed point is **unstable**; in a perfect world we could keep the pendulum upright but any small ‘nudge’ will result in the system changing.

Knowing the invariant solutions helps us to sketch the phase portrait without having to explicitly calculate individual orbits.



We can also find another type of invariant solution called **heteroclinic orbits** that connect stable and unstable fixed points. This gives us even more information of the phase portrait: With this information we now know everything about the dynamical system; the invariant solutions **order** the dynamics of the system; if the initial conditions are ‘inside’ the heteroclinic orbit then the pendulum is in libration, if the initial conditions are ‘outside’ then the pendulum is in rotation mode.



1.3.3 The dynamical systems algorithm

Definition 1.3.6: The algorithm (to be developed over the course)

Let an n -dynamical system be described by a set of ODE of the form

$$\dot{x} = f(x). \quad (1.3.16)$$

We will seek to understand the dynamics of this system without explicitly solving it (or by a computer). We will do this using the following steps

- Step 1: Find invariant solutions
- Step 2: Determine whether they are stable or unstable
- Step 3: Complete the phase portrait
- Step 4: **Determine/understand how a system evolves**

Remark 1. *In dynamical systems theory we are not interested in solution methods to find explicit solutions; rather we want to glean information about the possible end states of the system without explicitly solving the system.*

Remark 2. *This ‘algorithm’ can be applied regardless of the dimensionality of the system. This is one of the chief strengths/beauties of dynamical systems theory.*

1.3.4 Formal definitions

Before we continue it is worthwhile giving a formal definition of a dynamical system. It is important to note that ODEs are only one way we can define a dynamical system; we could define them by recursion relations or simply a set of experimental observations. We need a definition that encompasses all of these representations.

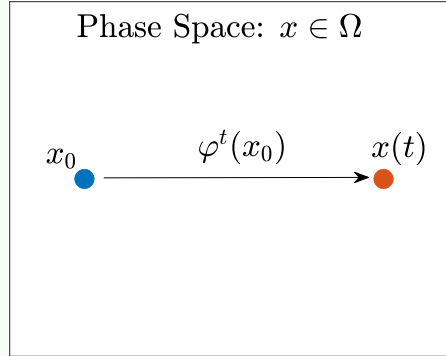
Definition 1.3.7: A dynamical system

A dynamical system is a triple $\{T, \Omega, \varphi^t\}$, where T represents a time set and $\varphi^t : \Omega \mapsto \Omega$ is a family of operation operators parameterised by $t \in T$ and satisfying the properties:

$$\varphi^0(x) = x, \quad \forall x \in \Omega \quad (1.3.17)$$

$$\varphi^t(\varphi^s(x)) = \varphi^{t+s}(x), \quad \forall x \in \Omega, t, s \in T \quad (1.3.18)$$

This is shown schematically in the figures below:



Remark 3. In this course we will only concentrate on dynamical systems described by ODEs, which are sometimes called **flows**

Definition 1.3.8: A dynamical system (flow)

A dynamical system (flow) is an n -dimensional ODE of the form

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad (1.3.19)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state space, $t \in \mathbb{R}$, $\mathbf{f} : \mathbb{R}^n \mapsto \mathbb{R}^n$ and $\varphi^t(x)$ is defined by the ODE.

Chapter 2

One-dimensional dynamical systems

In this chapter, we'll discuss dynamical systems of one dimension, also referred to as dynamical systems on a line. These are characterised by the equation

$$\dot{x} = f(x). \quad (2.0.1)$$

We should remember that the general solution of any first order ODE (before imposing any initial or boundary conditions) is defined up to an arbitrary constant of integration. In fact, if we have a system of n first order ODEs, then we obtain n arbitrary constants, and thus require n initial or boundary conditions in order to obtain a unique solution to the system. Imposing a particular condition (or set of conditions) on our problem(s) thus impacts what our solution will look like. The general solution of the system can be thought of as representing all possible solutions to the problem. Let us consider the following example to help us understand the approach taken in this chapter and course.

2.1. Stability of fixed-points

The last lecture discussed the importance of finding invariant solutions of a dynamical system and finding out whether they are stable or not so we can plot the phase portrait and understand the complete phase-portrait. The simplest form of invariant solution is a fixed-point which as we stated in the previous chapter is defined as those values of $x = x_e$ in (2.0.1) for which $\dot{x} = 0$ and hence

$$f(x_e) = 0 \quad (2.1.1)$$

Calculation 2.1.1: Taylor expansions

The next task is to understand whether the fixed-points are a long time dynamical outcome of the system. Let's imagine that at a time t we are close to the fixed point, x_e , i.e.

$$x(t) = x_e + \delta v(t), \quad \delta \ll 1. \quad (2.1.2)$$

Inserting this into the dynamical system

$$\dot{x} = f(x), \quad (2.1.3)$$

gives

$$\delta \dot{v} = f(x_e + \delta v). \quad (2.1.4)$$

We now Taylor expand the RHS so that

$$\delta \dot{v} = \underbrace{f(x_e)}_{=0} + \delta f'(x_e)v + O(\delta^2), \quad (2.1.5)$$

so that if we ignore higher powers of δ because they are very small we get

$$\dot{v} = f'(x_e)v. \quad (2.1.6)$$

This ODE is linear and can be solved to give

$$v(t) = Ae^{f'(x_e)t}, \quad (2.1.7)$$

where A is an arbitrary constant. Now it is clear that if $f'(x_e) < 0$ then $v(t) \rightarrow 0$ as $t \rightarrow \infty$ but if $f'(x_e) > 0$ then $v(t) \rightarrow \infty$ as $t \rightarrow \infty$. Therefore we arrive at the following theorem

Theorem 2.1.1: Linear stability of a one-dimensional fixed point

A fixed point x_e of the one-dimensional dynamical system

$$\dot{x} = f(x) \quad (2.1.8)$$

is said to be **linearly stable** if

$$f'(x_e) < 0, \quad (2.1.9)$$

linearly unstable if

$$f'(x_e) > 0 \quad (2.1.10)$$

and **neutrally stable** if

$$f'(x_e) = 0 \quad (2.1.11)$$

Remark 4. If a fixed point is linearly unstable then it is also asymptotically unstable. However a fixed point can be linearly stable and not be asymptotically stable; we will examine examples of this later on in the course.

Example 2.1.1

Consider the one-dimensional ODE for $x \in \mathbb{R}$

$$\dot{x} = x(x+1)(x-2). \quad (2.1.12)$$

Complete the global phase portrait without explicitly solving the ODE.

Solution 2.1.1

Step 1: Locate the fixed points. In this example

$$f(x) = x(x+1)(x-2) = x(x^2 - x - 2) = x^3 - x^2 - 2x \quad (2.1.13)$$

Fixed points occur when $\dot{x} = 0$, i.e.

$$x(x+1)(x-2) = 0 \implies x = -1, 0, 2. \quad (2.1.14)$$

Step 2: Determine the stability. To determine the stability of the fixed points we need to find $f'(x)$ which is

$$f'(x) = 3x^2 - 2x - 2. \quad (2.1.15)$$

Now at the fixed point $x_e = -1$ we have

$$f'(-1) = 3 + 2 - 2 = 3 > 0, \quad (2.1.16)$$

and hence $x_e = -1$ is unstable; initial conditions nearby will diverge away from $x = -1$. At the fixed point $x_e = 0$ we have

$$f'(0) = -2 < 0, \quad (2.1.17)$$

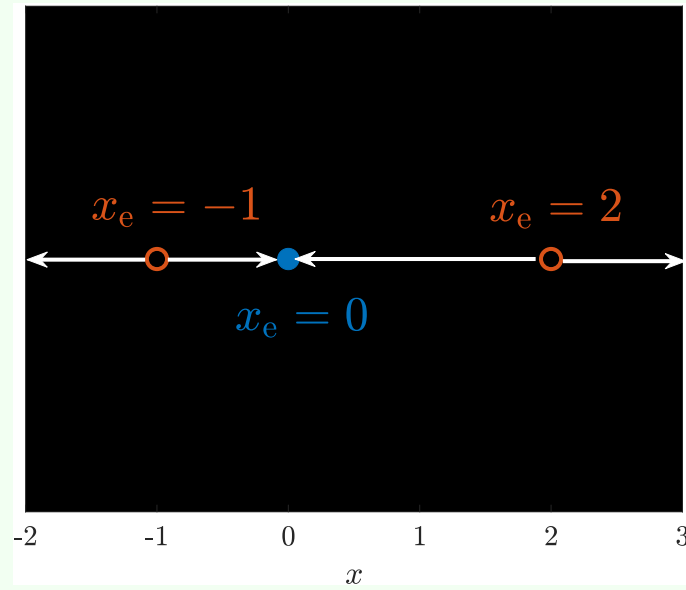
and hence $x_e = 0$ is stable; initial conditions nearby will tend to $x = 0$. Finally at the fixed point $x_e = 2$ we have

$$f'(2) = 12 - 4 - 2 = 4 > 0, \quad (2.1.18)$$

and hence $x_e = 2$ is unstable; initial conditions nearby will diverge from $x = 2$.

Step 3: Sketch the phase-portrait. Now we have all the information we need to sketch the phase-portrait. Notice we have not gone through the pain of explicitly solving the ODE (which of course we can if we can be bothered).

The phase-space is just the real-line. We usually mark stable fixed points with closed circles and unstable fixed points as open circles. With this knowledge we can draw arrows indicating where $x(t)$ will evolve depending on its initial state.

**Example 2.1.2**

Consider the one-dimensional ODE for $x \in \mathbb{R}$

$$\dot{x} = \sinh(x^2 - 4x + 3) \quad (2.1.19)$$

Solution 2.1.2

Complete the global phase portrait without explicitly solving the ODE.

Step 1: Locate the fixed points. In this example

$$f(x) = \sinh(x^2 - 4x + 3) \quad (2.1.20)$$

Fixed points occur when $\dot{x} = 0$, i.e.

$$(x - 1)(x - 3) = 0 \implies x = 1, 3. \quad (2.1.21)$$

Step 2: Determine the stability. To determine the stability of the fixed points we need to find $f'(x)$ which is

$$f'(x) = 2(x - 2) \cosh(x^2 - 4x + 3) \quad (2.1.22)$$

Now at the fixed point $x_e = 1$ we have

$$f'(1) = -2 \cosh(0) = -2 < 0 \quad (2.1.23)$$

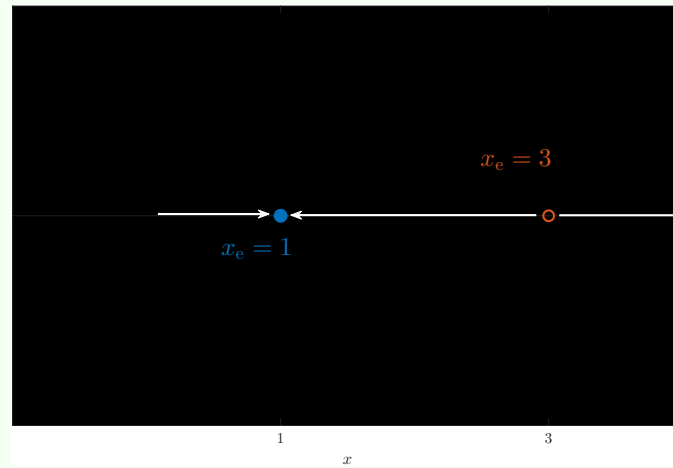
and hence $x_e = 1$ is stable; initial conditions nearby will converge away from $x = 1$. At the fixed point $x_e = 3$ we have

$$f'(3) = 2 \cosh(0) = 2 > 0, \quad (2.1.24)$$

and hence $x_e = 3$ is unstable; initial conditions nearby will diverge away from $x = 0$.

Step 3: Sketch the phase-portrait. Now we have all the information we need to sketch the phase-portrait. Notice we have not gone through the pain of explicitly solving the ODE (which of course we can if we can be bothered).

The phase-space is just the real-line. We usually mark stable fixed points with closed circles and unstable fixed points as open circles. With this knowledge we can draw arrows indicating where $x(t)$ will evolve depending on it's initial state.



Sometimes this method fails when the fixed-point is **non-hyperbolic** (this definition will be made clearer later in the course). In this case we have to consider the direction of the flow.

Example 2.1.3

Consider the one-dimensional ODE for $x \in \mathbb{R}$

$$\dot{x} = x^2(1 - x) \quad (2.1.25)$$

Solution 2.1.3

Complete the global phase portrait without explicitly solving the ODE.

Step 1: Locate the fixed points. In this example

$$f(x) = x^2(1 - x) \quad (2.1.26)$$

Fixed points occur when $\dot{x} = 0$, i.e.

$$x^2(1 - x) = 0 \implies x = 0, 1. \quad (2.1.27)$$

Step 2: Determine the stability. To determine the stability of the fixed points we need to find $f'(x)$ which is

$$f'(x) = 2x - 3x^2 \quad (2.1.28)$$

Now at the fixed point $x_e = 1$ we have

$$f'(1) = 2 - 3 = -1 < 0 \quad (2.1.29)$$

and hence $x_e = 1$ is stable; initial conditions nearby will converge away from $x = 1$. However, at the fixed point $x_e = 0$ we have

$$f'(0) = 0. \quad (2.1.30)$$

This is a problem, this means that the fixed-point is what we call **non-hyperbolic** and we say that the fixed-point is **neutrally stable**.

To understand the stability we have to consider the ‘flow’ of the ODE near the fixed-point. We do this by examining the sign of $\dot{x}$ near the non-hyperbolic fixed point $x = 0$.

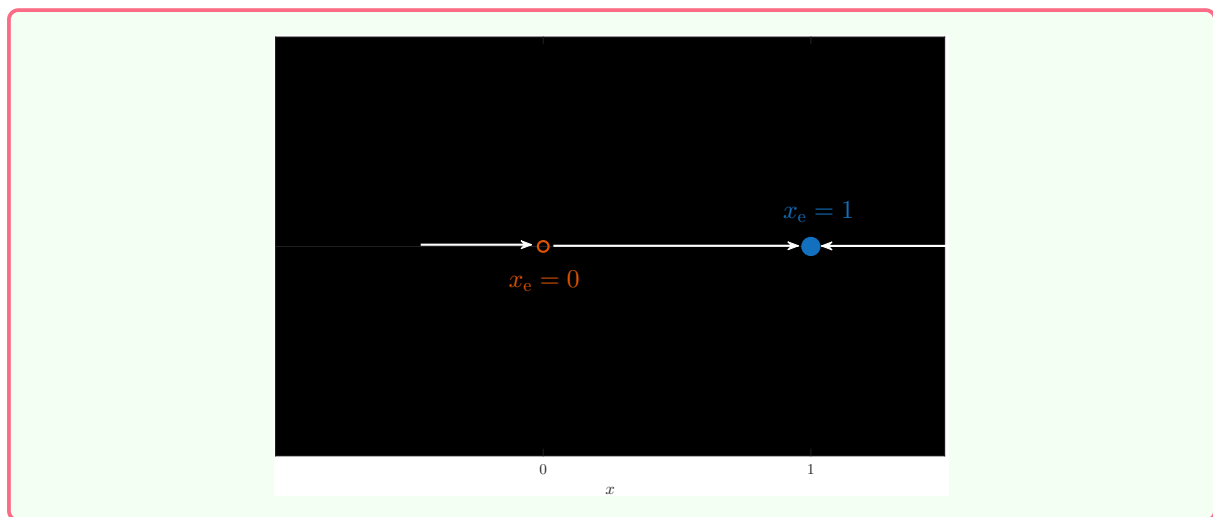
When x is close to 0, $f(x) > 0$. This means that

$$\dot{x} > 0, \quad (2.1.31)$$

if x is close to 0. This means x increase as t increases near $x = 0$. Initial conditions less than 0 will converge to $x = 0$ but initial conditions larger than 0 will diverge. The fixed-point is neutrally stable or marginally stable.

Step 3: Sketch the phase-portrait. Now we have all the information we need to sketch the phase-portrait. Notice we have not gone through the pain of explicitly solving the ODE (which of course we can if we can be bothered).

The phase-space is just the real-line. We usually mark stable fixed points with closed circles and unstable fixed points as open circles. With this knowledge we can draw arrows indicating where $x(t)$ will evolve depending on it’s initial state.



Chapter 3

One-dimensional bifurcations

As we've shown so far, the behaviour of solutions to dynamical systems on the line is very limited - solutions can only tend to $\pm\infty$ or to an equilibrium point. So what makes them interesting? The answer is their dependence on parameters. As we vary parameters in our problem, the qualitative structure of our solutions can change. Such changes in structure are called bifurcations, and the points at which they occur are known as bifurcation points.

Bifurcations are important when modelling physical systems, they represent the transition between states as some control parameter is varied. Consider, for example, the application of force to a beam, if the force applied is too large, the beam will buckle. The bifurcation here is the change in shape of the beam, the control parameter in this problem is the force applied, and the bifurcation point is the force applied in the instant that the beam begins to buckle.

The number and type of invariant solutions may vary when the fixed parameter is varied. When the global phase portrait drastically changes due to a parameter approaching a distinct value we say that a **bifurcation** has occurred. In this lecture we will examine a number of fundamental one-dimensional bifurcations and sketch **bifurcation diagrams** which encodes this information succinctly.

As in the last chapter, we'll discuss dynamical systems of one dimension, also referred to as dynamical systems on a line. Now we will let the dynamical system depend on a fixed parameter, i.e.

$$\dot{x} = f(x; \alpha). \quad (3.0.1)$$

We will discuss three types of bifurcations through three examples

Example 3.0.1: Fold/Saddle-node bifurcations

Saddle-node bifurcations are the basic mechanism by which equilibrium points are created or destroyed. Consider the dynamical system

$$\dot{x} = \alpha - x^2, \quad x \in \mathbb{R}, \quad (3.0.2)$$

where $\alpha \in \mathbb{R}$ is a fixed parameter.

1. Sketch the phase portrait for the system in the cases $\alpha < 0$, $\alpha = 0$ and $\alpha > 0$.
2. Sketch a bifurcation diagram to show the number of types of solution as α is varied.

Solution 3.0.1: Fold/Saddle-node bifurcations

Fixed points occur when $\dot{x}$ and hence occur when

$$x = x_e = \pm\sqrt{\alpha}. \quad (3.0.3)$$

Clearly, when $\alpha < 0$ there are no solutions (remember the state-space is $x \in \mathbb{R}$) and $\alpha > 0$ there are two solutions. Let's examine the stability of these solutions.

First we calculate $f'(x_e)$ which is

$$f'(x) = -2x, \quad (3.0.4)$$

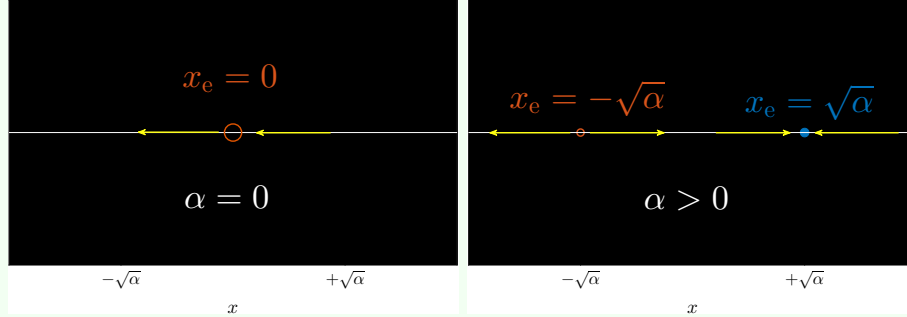
hence the fixed point at $x_e = +\sqrt{\alpha}$ is **stable** and the fixed point at $x_e = -\sqrt{\alpha}$ is unstable. The global phase portrait is shown in figure below

What happens when $\alpha = 0$? There is only one fixed-point, $x_e = 0$ and at this point $f'(x_e) = 0$ and our linear theory is not useful to us. Indeed, this is an example of a **non-hyperbolic** fixed-point

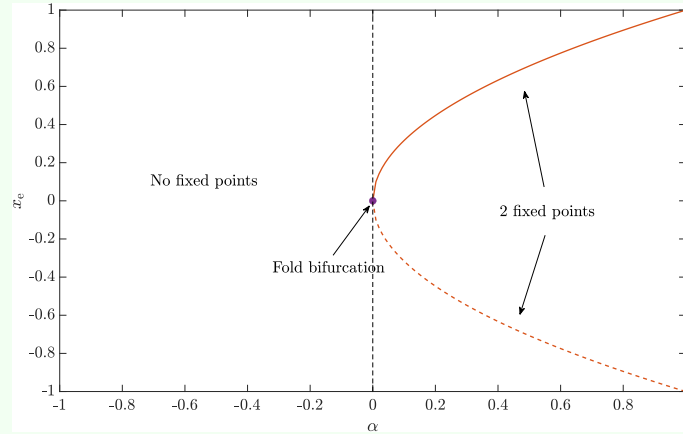
which requires more work to determine the stability. When $\alpha = 0$ and $x \neq 0$ we have

$$\dot{x} < 0, \quad \forall x \neq 0. \quad (3.0.5)$$

This means that that x **decreases** as t **increases** and hence trajectories that have an initial condition $x_0 > 0$ converge to $x_e = 0$ whilst if the initial condition, $x_0 < 0$, the trajectory will diverge as shown in the examples below:



We can now put all of this information together in a **bifurcation diagram**. This is a plot of the fixed-points, x_e against the parameter α . If x_e is stable we mark the curve as a solid line and if it is unstable we mark it as a dashed line as shown in the figure below:



Example 3.0.2: Transcritical bifurcations

Some models require that we always have at least one equilibrium point which can never be destroyed. As an example, any attempt to model population growth must always have an equilibrium point when the population is zero! However these 'fixed' equilibrium points can undergo a change of stability, the basic mechanism for such a change is a transcritical bifurcation.

Consider the dynamical system

$$\dot{x} = x(\alpha - x), \quad x \in \mathbb{R}, \quad (3.0.6)$$

where $\alpha \in \mathbb{R}$ is a fixed parameter.

1. Sketch the phase portrait for the system in the cases $\alpha < 0$, $\alpha = 0$ and $\alpha > 0$.
2. Sketch a bifurcation diagram to show the number of types of solution as α is varied.

Solution 3.0.2: Transcritical bifurcations

Fixed points occur when $\dot{x}$ and hence occur when

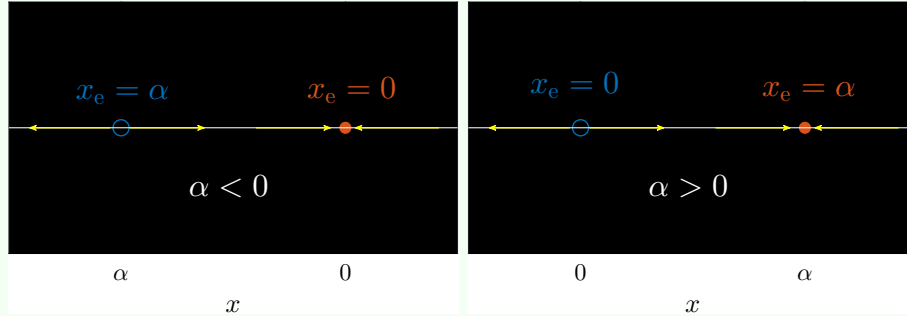
$$x = x_e = 0, \alpha. \quad (3.0.7)$$

Clearly, when $\alpha = 0$ there is only one solution at the origin. Let's examine the stability of these solutions.

First we calculate $f'(x_e)$ which is

$$f'(x) = \alpha - 2x, \quad (3.0.8)$$

hence if $\alpha > 0$, the fixed point at $x_e = 0$ is **unstable** and the fixed point at $x_e = \alpha$ is **stable**. When $\alpha < 0$, the fixed point at $x_e = 0$ is **stable** and the fixed point at $x_e = \alpha$ is **unstable**; at $\alpha = 0$ the stability properties of the two fixed-points ‘switch’; see figures below.

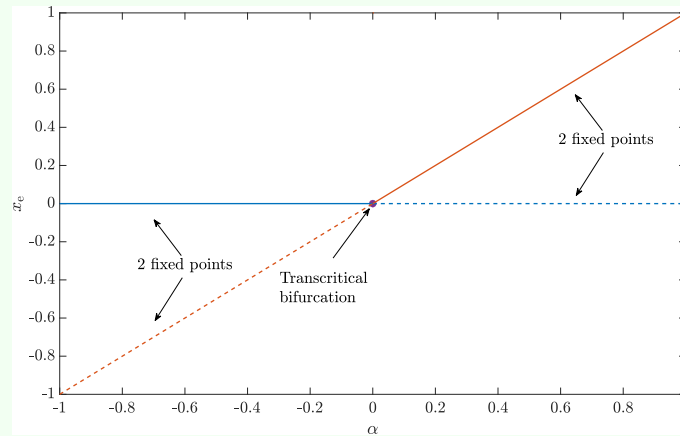


What happens when $\alpha = 0$? There is only one fixed-point, $x_e = 0$ and at this point $f'(x_e) = 0$ and our linear theory is not useful to us. Indeed, this is an example of a **non-hyperbolic** fixed-point which requires more work to determine the stability. When $\alpha = 0$ and $x \neq 0$ we have

$$\dot{x} < 0, \quad \forall x \neq 0. \quad (3.0.9)$$

This means that that x **decreases** as t **increases** and hence $x_e = 0$ is marginally stable as trajectories to the right converge to the fixed-point and trajectories to the left diverge from the fixed-point.

We can now put all of this information together in a **bifurcation diagram**. This is a plot of the fixed-points, x_e against the parameter α . If x_e is stable we mark the curve as a solid line and if it is unstable we mark it as a dashed line as shown in the figure below. Notice how the marking of the solution curves ‘switch’ as they go through the origin. When $\alpha = 0$ we say that there is a **transcritical bifurcation** as shown in the figure below:



The final type of bifurcation we will deal with is the pitchfork bifurcation, which usually occurs in problems where there is some symmetry. For example, many problems have some spatial symmetry between left and right, and in these cases equilibrium points are often created and destroyed in symmetrical pairs. We will consider two types of pitchfork bifurcation, beginning with the more simple case supercritical pitchfork bifurcation.

Example 3.0.3: Supercritical pitchfork bifurcations

Consider the dynamical system

$$\dot{x} = x(\alpha - x^2), \quad x \in \mathbb{R}, \quad (3.0.10)$$

where $\alpha \in \mathbb{R}$ is a fixed parameter.

1. Sketch the phase portrait for the system in the cases $\alpha < 0$, $\alpha = 0$ and $\alpha > 0$.

2. Sketch a bifurcation diagram to show the number of types of solution as α is varied.

Solution 3.0.3: Supercritical pitchfork bifurcations

Fixed points occur when $\dot{x}$ and hence occur when

$$x = x_e = 0, \pm\sqrt{\alpha}. \quad (3.0.11)$$

Clearly, when $\alpha \leq 0$ there is only one solution. Let's examine the stability of these solutions.

First we calculate $f'(x_e)$ which is

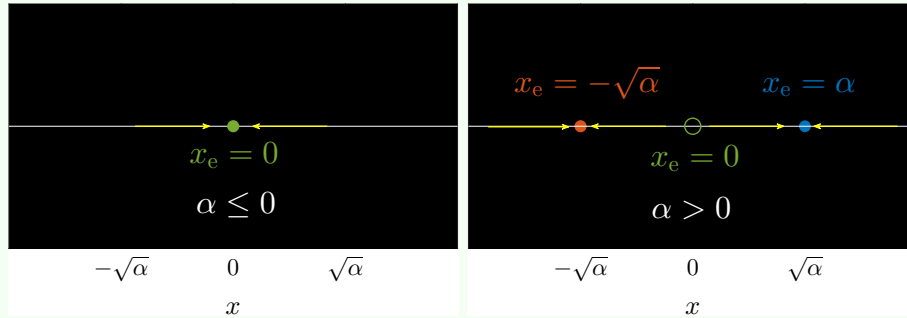
$$f'(x) = \alpha - 3x^2, \quad (3.0.12)$$

hence if $\alpha > 0$, the fixed point at $x_e = 0$ is **unstable**, the fixed points at $x_e = -\pm\sqrt{\alpha}$ are both **stable**, as shown in the figure below. If $\alpha < 0$ then the (only) fixed point at $x_e = 0$ is **stable**.

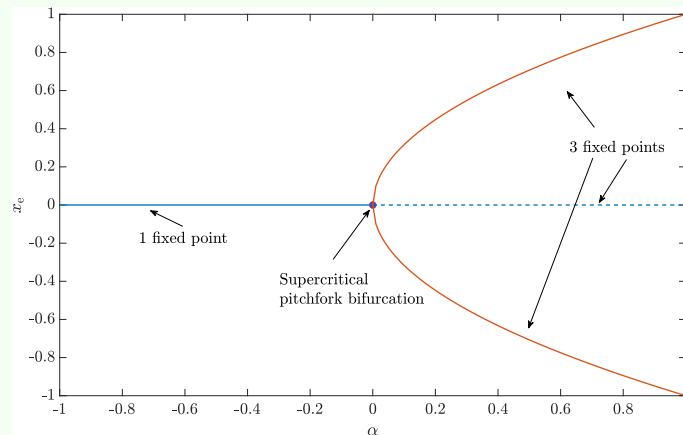
What happens when $\alpha = 0$? There is only one fixed-point, $x_e = 0$ and at this point $f'(x_e) = 0$ and our linear theory is not useful to us. Indeed, this is an example of a **non-hyperbolic** fixed-point which requires more work to determine the stability. When $\alpha = 0$ and $x \neq 0$ we have

$$\dot{x} < 0, \quad \forall x > 0, \quad \text{and} \quad \dot{x} > 0, \quad \forall x < 0, \quad (3.0.13)$$

This means that that for $x > 0$, x **decreases** as t **increases** but for $x < 0$, x **increases** as t **increase**. Hence, despite being non-hyperbolic, $x_e = 0$ is stable as shown in the figures below;



We can now put all of this information together in a **bifurcation diagram**. This is a plot of the fixed-points, x_e against the parameter α . If x_e is stable we mark the curve as a solid line and if it is unstable we mark it as a dashed line as shown in the figure below. Notice how the diagram for $\alpha > 0$ resembles a 'pitchfork'. Hence, when $\alpha = 0$ we say that there is a **supercritical pitchfork bifurcation**.



Example 3.0.4: Subcritical pitchfork bifurcations

Consider the dynamical system

$$\dot{x} = x(\alpha + x^2), \quad x \in \mathbb{R}, \quad (3.0.14)$$

where $\alpha \in \mathbb{R}$ is a fixed parameter.

1. Sketch the phase portrait for the system in the cases $\alpha < 0$, $\alpha = 0$ and $\alpha > 0$.
2. Sketch a bifurcation diagram to show the number of types of solution as α is varied.

Solution 3.0.4: Subcritical pitchfork bifurcations

Fixed points occur when $\dot{x}$ and hence occur when

$$x = x_e = 0, \pm\sqrt{-\alpha}. \quad (3.0.15)$$

Clearly, when $\alpha \geq 0$ there is only one solution. Let's examine the stability of these solutions. First we calculate $f'(x_e)$ which is

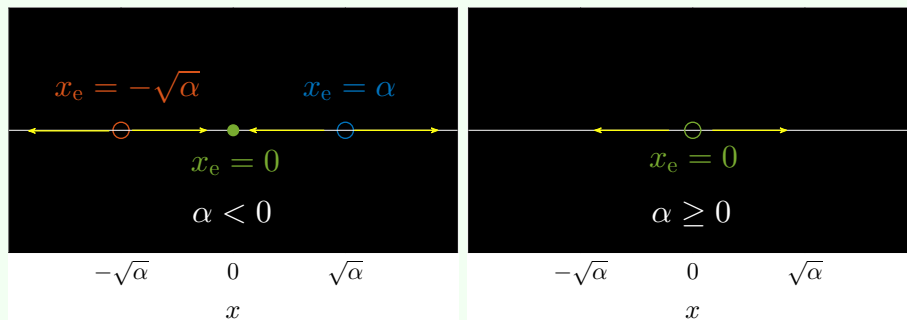
$$f'(x) = \alpha + 3x^2, \quad (3.0.16)$$

hence if $\alpha < 0$, the fixed point at $x_e = 0$ is **stable**, the fixed points at $x_e = -\pm\sqrt{-\alpha}$ are both **unstable**, as shown in the figures below. If $\alpha > 0$ then the (only) fixed point at $x_e = 0$ is **unstable**.

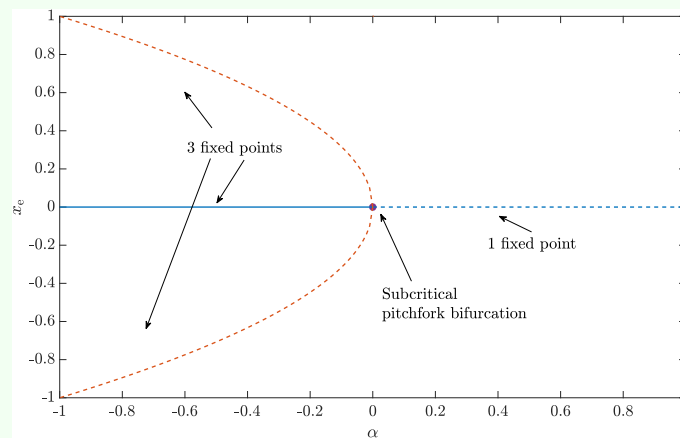
What happens when $\alpha = 0$? There is only one fixed-point, $x_e = 0$ and at this point $f'(x_e) = 0$ and our linear theory is not useful to us. Indeed, this is an example of a **non-hyperbolic** fixed-point which requires more work to determine the stability. When $\alpha = 0$ and $x \neq 0$ we have

$$\dot{x} > 0, \quad \forall x > 0, \quad \text{and} \quad \dot{x} < 0, \quad \forall x < 0, \quad (3.0.17)$$

This means that that for $x > 0$, x **decreases** as t **increases** but for $x < 0$, x **increases** as t **increase**. Hence, despite being non-hyperbolic, $x_e = 0$ is unstable, as shown in the figures below;



We can now put all of this information together in a **bifurcation diagram**. This is a plot of the fixed-points, x_e against the parameter α . If x_e is stable we mark the curve as a solid line and if it is unstable we mark it as a dashed line as shown in the figure below. Notice how the diagram for $\alpha < 0$ resembles a 'pitchfork'. Hence, when $\alpha = 0$ we say that there is a **subcritical pitchfork bifurcation**.



Chapter 4

Let's talk about stability

We now move on to a general n - dimensional dynamical system of the form:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n \quad (4.0.1)$$

In two (or more) dimension the concept of stability of a fixed-point becomes a little more nuanced. In fact, as we will now discuss, there are actually a number of different criteria for stability; that range from really strict (global) to weak (linear). Key to these criteria is the identification of a suitable norm defined on the phase space. In $\mathbb{R}^n$ this is usually defined as the Euclidean norm, i.e.

$$d(\mathbf{x}, \mathbf{y}) = |\mathbf{x}(t) - \mathbf{y}(t)| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots (x_n - y_n)^2}, \quad (4.0.2)$$

where $\mathbf{x} = [x_1, x_2, \cdots x_n]^T$ and $\mathbf{y} = [y_1, y_2, \cdots y_n]^T$.

4.1. Stability criteria

The strictest definition is **global** stability. This essentially states that

All initial conditions in phase-space converge to the fixed-point as $t \rightarrow \infty$.

The formal definition is:

Definition 4.1.1: Global stability

A fixed point $\mathbf{x}_e$ in the phase-space Ω , is said to be globally stable if $\forall \mathbf{x}(t=0) \in \Omega$

$$|\mathbf{x}(t) - \mathbf{x}_e| \rightarrow 0, \quad \text{as} \quad t \rightarrow \infty. \quad (4.1.1)$$

For an example of 1D dynamical system that has a globally stable fixed-point; see the lecture videos. The next strictest definition is **asymptotic** stability. This essentially states that

All initial conditions in a ball centred around the fixed-point converge to the fixed-point as $t \rightarrow \infty$.

The formal definition is:

Definition 4.1.2: Asymptotic stability

A fixed point $\mathbf{x}_e$ in the phase-space Ω , is said to be asymptotically stable if there exists a $\delta > 0$ such that $\forall \mathbf{x}(t=0) \in \Omega$ with

$$|\mathbf{x}(0) - \mathbf{x}_e| < \delta, \quad (4.1.2)$$

then

$$|\mathbf{x}(t) - \mathbf{x}_e| \rightarrow 0, \quad \text{as} \quad t \rightarrow \infty. \quad (4.1.3)$$

For an example of 1D dynamical system that has an asymptotic stable fixed-point; see the lectures. A slightly less strict definition is **Lyapunov** stability. This essentially states that

All initial conditions in a ball centred around the fixed-point never leave the ball as $t \rightarrow \infty$.

The formal definition is:

Definition 4.1.3: Lyapunov stability

A fixed point $\mathbf{x}_e$ in the phase-space Ω , is said to be Lyapunov stable if for all $\epsilon > 0$, there exists a $\delta > 0$ such that if

$$|\mathbf{x}(0) - \mathbf{x}_e| < \delta, \quad (4.1.4)$$

then

$$|\mathbf{x}(t) - \mathbf{x}_e| < \epsilon, \quad \forall t \geq 0. \quad (4.1.5)$$

This allows us to classify fixed points as stable if surrounded by periodic-orbits as we shall see later in dynamical systems in $\mathbb{R}^2$.

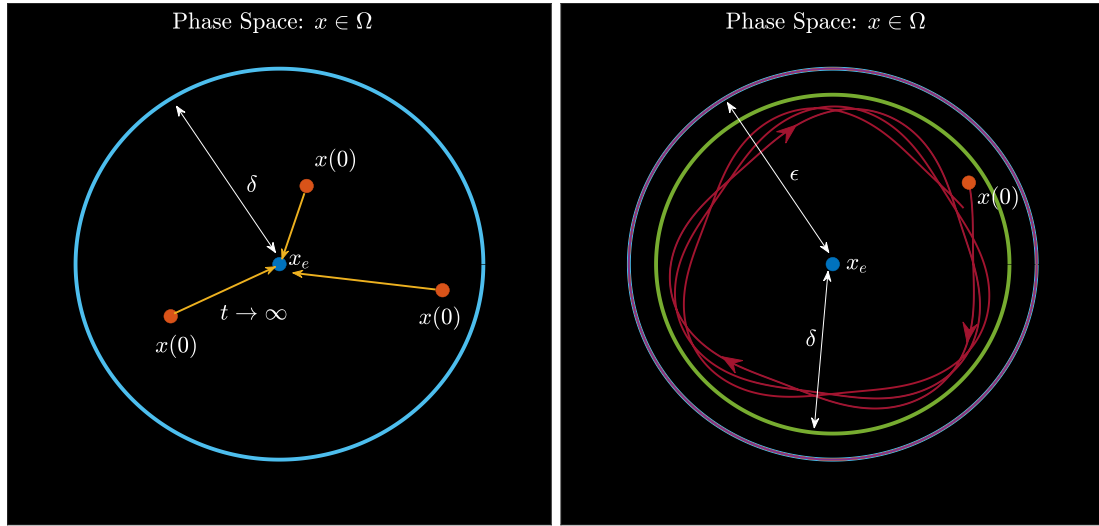


Figure 4.1: Two types of stability criteria. Left: asymptotic stability and Right: Lyapunov stability. Note that if a fixed point is asymptotically stable then it is also Lyapunov stable but the converse is not true

The weakest condition is **linear stability**. This essentially states that

All initial conditions in a small ball centred around the fixed-point converge to the fixed-point exponentially as $t \rightarrow \infty$.

The formal definition is:

Definition 4.1.4: Linear stability/exponential

A fixed point x_e is said to be linearly stable if all $\epsilon > 0$, there exists a δ and σ such that if

$$|\mathbf{x}(0) - \mathbf{x}_e| < \delta, \quad \text{as } t \rightarrow \infty. \quad (4.1.6)$$

then

$$|\mathbf{x}(t) - \mathbf{x}_e| \leq \epsilon e^{-\sigma t}, \quad (4.1.7)$$

4.2. Linear stability

Determining asymptotic/Lyapunov stability of a fixed-point of a general nonlinear system is difficult and in general has no set of clear instructions to achieve this; it is very much a ‘dark-art’. However, not all is lost because a lot of the time linear stability is all we require (we will state a theorem in a subsequent chapter). Linear/exponential stability can be determined by linearising the system. We do this by Taylor expanding dynamical-system about it’s fixed-point.

Calculation 4.2.1: Taylor expansions in n - dimensions

The next task is to understand whether the fixed-points are a long time dynamical outcome of the system. Let's imagine that at a time t we are close to the fixed point, $\mathbf{x}_e$, i.e.

$$\mathbf{x}(t) = \mathbf{x}_e + \delta \mathbf{v}(t), \quad \delta \ll 1. \quad (4.2.1)$$

Inserting this into the dynamical system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x})]^T. \quad (4.2.2)$$

gives

$$\delta \dot{\mathbf{v}} = \mathbf{f}(\mathbf{x}_e + \delta \mathbf{v}). \quad (4.2.3)$$

We now Taylor expand the RHS so that

$$\delta \dot{\mathbf{v}} = \underbrace{\mathbf{f}(\mathbf{x}_e)}_{=0} + \delta \mathbf{J}(\mathbf{x}_e) \mathbf{v} + O(\delta^2), \quad (4.2.4)$$

where $\mathbf{J}$ is the Jacobian matrix defined as

$$\mathbf{J} = J_{ij} = \frac{\partial f_i}{\partial x_j}, \quad \text{with} \quad \mathbf{x} = [x_1, x_2, \dots, x_n]^T. \quad (4.2.5)$$

If we ignore higher powers of δ because they are very small we get

$$\dot{\mathbf{v}} = \mathbf{J}(\mathbf{x}_e) \mathbf{v}. \quad (4.2.6)$$

We now assume a solution of the form

$$\mathbf{v} \mapsto \mathbf{v} e^{\sigma t}. \quad (4.2.7)$$

Therefore

$$\sigma \mathbf{v} = \mathbf{J} \mathbf{v}. \quad (4.2.8)$$

This is an eigenvalue problem to solve for σ and $\mathbf{v}$. The eigenvalues satisfy

$$\det(\sigma \mathbf{I} - \mathbf{J}) = 0, \quad (4.2.9)$$

where $\mathbf{I}$ is the identity. The nature of the eigenvalues, σ , which we call the **spectrum**, determines the stability.

Theorem 4.2.1: Linear stability of an n -dimensional fixed point

Let $\sigma(\mathbf{x}_e)$ be the spectrum of a fixed point $\mathbf{x}_e$ of the one-dimensional dynamical system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}). \quad (4.2.10)$$

Let n^+ be the number of eigenvalues with $\text{Re}(\sigma) > 0$, n^- be the number of eigenvalues with $\text{Re}(\sigma) < 0$ and n^0 be the number of eigenvalues where $\text{Re}(\sigma) = 0$.

The fixed-point $\mathbf{x}_e$ is said to be **linearly stable** if

$$n^+ = n^0 = 0, \quad (4.2.11)$$

linearly unstable if

$$n^+ > 0 \quad (4.2.12)$$

and **neutrally stable** if

$$n^- > 0. \quad (4.2.13)$$

4.3. Hierarchy of stability criteria

We have established a hierarchy of stability criteria. The following logical implication is true

$$\text{Linear instability} \implies \text{Lyapunov instability} \implies \text{Asymptotic instability} \implies \text{Global instability}. \quad (4.3.1)$$

Stability of a fixed-point is slightly more nuanced. However, there is a theorem, due to Lyapunov that states

Theorem 4.3.1: Lyapunov (1892)

A fixed-point is asymptotically stable (and hence Lyapunov stable) if it is linearly stable if $\mathbf{f}(\mathbf{x}_e)$ is sufficiently smooth (i.e. enough derivatives exist).

In the next chapter we show this explicitly by looking at linear systems in $\mathbb{R}^2$.

Chapter 5

Linear two-dimensional dynamical systems

We now move on to linear dynamical systems on the plane.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^2 \quad (5.0.1)$$

where $\mathbf{A}$ is a 2×2 matrix. In this lecture we are going to identify all of the types of fixed points that can exist in the system above.

5.1. Classification of all types of linear fixed points in $\mathbb{R}^2$

We will now discuss all possible types of fixed points of a **linear two-dimensional** system and sketch their phase portraits; these are fundamental objects and concepts in dynamical systems theory and require special and careful attention as these concepts can be extended to general n -dimensional systems.

Example 5.1.1: Stable nodes

Consider the dynamical system

$$\dot{x} = -5x + y \quad (5.1.1)$$

$$\dot{y} = x - 5y \quad (5.1.2)$$

1. Find the fixed point(s) of the system
2. Determine the stability of the fixed points(s)
3. Sketch the phase portrait.

Solution 5.1.1: Stable nodes

First we write it in matrix form, i.e.

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} -5 & 1 \\ 1 & -5 \end{pmatrix}. \quad (5.1.3)$$

There is only one fixed-point (check!) when $\dot{\mathbf{x}} = 0$ which is at the origin, i.e. $\mathbf{x}_e = \mathbf{0}$. To determine the stability of this solution, let's assume a solution of the form

$$\mathbf{x} = \mathbf{x}_e + \mathbf{v}e^{\sigma t}. \quad (5.1.4)$$

Substituting this into (5.1.20) gives

$$\sigma \mathbf{v} = \mathbf{A}\mathbf{v}, \quad (5.1.5)$$

which can be rearranged to

$$(\sigma \mathbf{I} - \mathbf{A}) \mathbf{v} = 0. \quad (5.1.6)$$

Remember your knowledge of linear algebra! For there to be a non-trivial $\mathbf{v}$ (i.e. not equal to the zero vector) to satisfy (5.1.6), we must have that

$$\det(\sigma \mathbf{I} - \mathbf{A}) = 0. \quad (5.1.7)$$

This is simply an eigenvalue problem. We can find the characteristic equation that σ satisfies, i.e.

$$(\sigma + 5)^2 - 1 = 0, \quad (5.1.8)$$

which has solutions

$$\sigma = -4, -6. \quad (5.1.9)$$

For $\sigma = -4$ we can find the eigenvector $\mathbf{v}$:

$$\mathbf{A}\mathbf{v} = -4\mathbf{v} \quad (5.1.10)$$

$$\Rightarrow \begin{pmatrix} -5u + v \\ u - 5v \end{pmatrix} = \begin{pmatrix} -4u \\ -4v \end{pmatrix} \quad (5.1.11)$$

$$\Rightarrow \begin{pmatrix} v - u \\ u - v \end{pmatrix} = \mathbf{0}, \quad (5.1.12)$$

so the eigenvector is any multiple of

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad (\sigma = -4). \quad (5.1.13)$$

Similarly the eigenvector for the eigenvalue $\sigma = -6$ can be calculated as

$$\mathbf{A}\mathbf{v} = -6\mathbf{v} \quad (5.1.14)$$

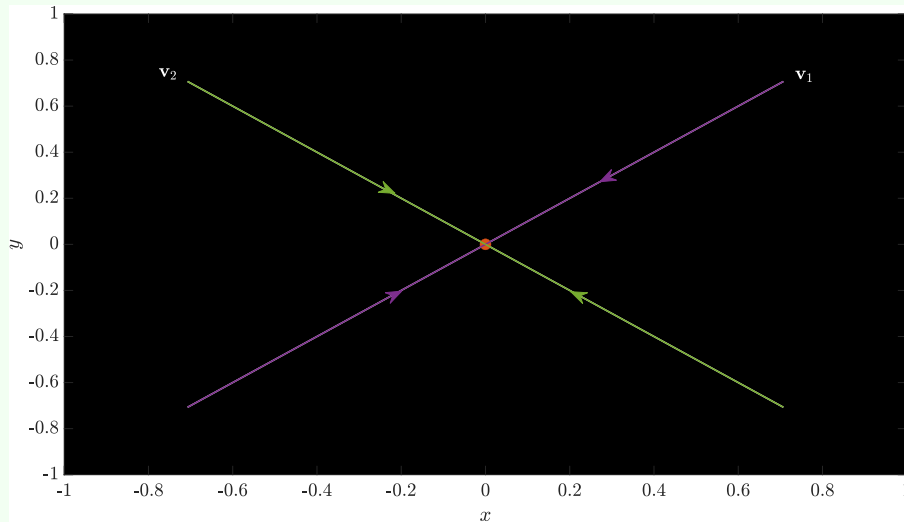
$$\Rightarrow \begin{pmatrix} -5u + v \\ u - 5v \end{pmatrix} = \begin{pmatrix} -6u \\ -6v \end{pmatrix} \quad (5.1.15)$$

$$\Rightarrow \begin{pmatrix} v + u \\ u + v \end{pmatrix} = \mathbf{0}, \quad (5.1.16)$$

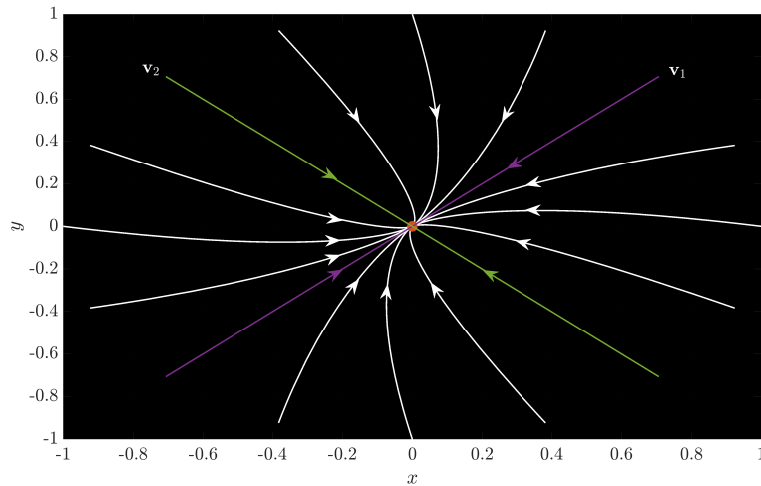
so the eigenvector is any multiple of

$$\mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad (\sigma = -6). \quad (5.1.17)$$

As both of the eigenvalues are negative then the fixed point at the origin is **stable** and is an example of a **stable node**. The eigenvectors are straight lines in the phase-plane as can be seen in the figure below:



Of the two eigenvalues, the eigenvalue which is **larger** is the dominant direction as $e^{-6t} < e^{-4t}$ for all t . Therefore all of the trajectories will enter the fixed-point in a direction parallel to the eigenvector $\mathbf{v}_1$ as shown in the figure below:

**Example 5.1.2: Unstable nodes**

Consider the dynamical system

$$\dot{x} = 5x - y \quad (5.1.18)$$

$$\dot{y} = -x + 5y \quad (5.1.19)$$

1. Find the fixed point(s) of the system
2. Determine the stability of the fixed points(s)
3. Sketch the phase portrait.

Solution 5.1.2: Unstable nodes

First we write it in matrix form, i.e.

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix}. \quad (5.1.20)$$

There is only one fixed-point (check!) when $\dot{\mathbf{x}} = 0$ which is at the origin, i.e. $\mathbf{x}_e = \mathbf{0}$. To determine the stability of this solution, let's assume a solution of the form

$$\mathbf{x} = \mathbf{x}_e + \mathbf{v}e^{\sigma t}. \quad (5.1.21)$$

Eventually we arrive at the characteristic equation that σ satisfies, i.e.

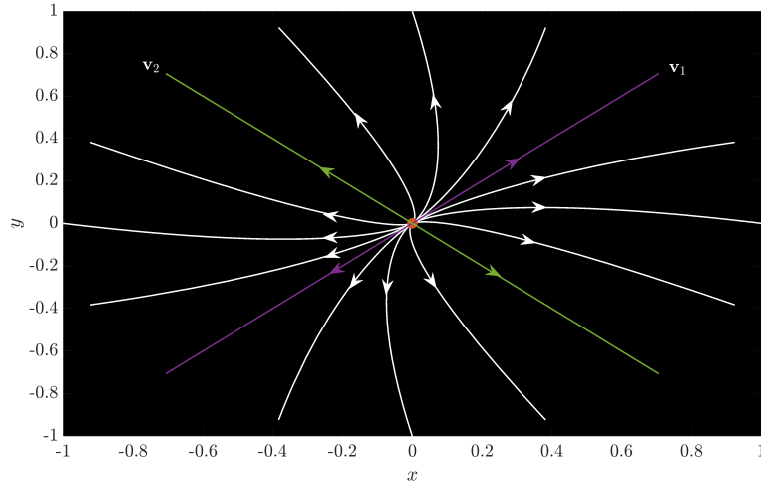
$$(\sigma - 5)^2 - 1 = 0, \quad (5.1.22)$$

with eigenvectors

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad (\sigma = 4). \quad (5.1.23)$$

$$\mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad (\sigma = 6). \quad (5.1.24)$$

As both of the eigenvalues are positive then the fixed point at the origin is **unstable** and is an example of a **unstable node**. The eigenvectors are straight lines in the phase-plane and the dominant eigenmode is $\mathbf{v}_1$ and hence trajectories enter the fixed-point parallel to this eigenmode as shown in the figure below:



Example 5.1.3: Stable spirals

Consider the dynamical system

$$\dot{x} = -x - 3y \quad (5.1.25)$$

$$\dot{y} = x - 2y \quad (5.1.26)$$

1. Find the fixed point(s) of the system
2. Determine the stability of the fixed points(s)
3. Sketch the phase portrait.

Solution 5.1.3: Stable spirals

First we write it in matrix form, i.e.

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} -1 & -3 \\ 1 & -2 \end{pmatrix}. \quad (5.1.27)$$

There is only one fixed-point (check!) when $\dot{\mathbf{x}} = \mathbf{0}$ which is at the origin, i.e. $\mathbf{x}_e = \mathbf{0}$.

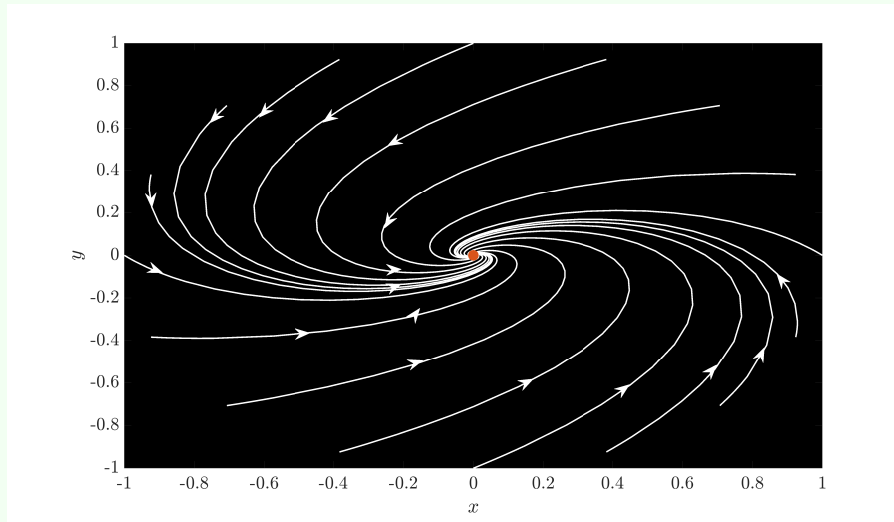
The eigenvalues (check!) are

$$\sigma = -\frac{3}{2} \pm \frac{\sqrt{11}}{2}i, \quad (5.1.28)$$

with solutions of the form

$$\mathbf{x} = A\mathbf{v}_1 e^{-\frac{3}{2}t} e^{\frac{\sqrt{11}}{2}it} + B\mathbf{v}_2 e^{-\frac{3}{2}t} e^{-\frac{\sqrt{11}}{2}it}, \quad (5.1.29)$$

where A and B are arbitrary. These solutions are oscillatory and decay as $t \rightarrow \infty$ hence this type of fixed-point is called a **stable spiral** as trajectories ‘spiral’ into the fixed-point, as shown in the figure below:

**Example 5.1.4: Unstable spiral**

Consider the dynamical system

$$\dot{x} = x - 3y \quad (5.1.30)$$

$$\dot{y} = x + 2y \quad (5.1.31)$$

1. Find the fixed point(s) of the system
2. Determine the stability of the fixed points(s)
3. Sketch the phase portrait.

Solution 5.1.4: Unstable spirals

First we write it in matrix form, i.e.

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}. \quad (5.1.32)$$

There is only one fixed-point (check!) when $\dot{\mathbf{x}} = \mathbf{0}$ which is at the origin, i.e. $\mathbf{x}_e = \mathbf{0}$.

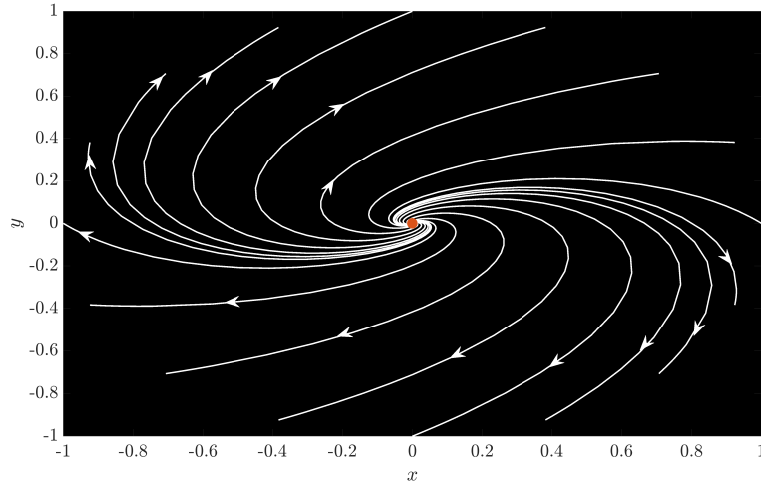
The eigenvalues (check!) are

$$\sigma = +\frac{3}{2} \pm \frac{\sqrt{11}}{2}i, \quad (5.1.33)$$

with solutions of the form

$$\mathbf{x} = A\mathbf{v}_1 e^{\frac{3}{2}t} e^{\frac{\sqrt{11}}{2}it} + B\mathbf{v}_2 e^{\frac{3}{2}t} e^{-\frac{\sqrt{11}}{2}it}, \quad (5.1.34)$$

where A and B are arbitrary. These solutions are oscillatory and blow-up as $t \rightarrow \infty$ hence this type of fixed-point is called a **stable spiral** as trajectories ‘spiral’ out of the fixed-point, as shown in the figure below:



Example 5.1.5: Saddle node

Consider the dynamical system

$$\dot{x} = -x + 5y \quad (5.1.35)$$

$$\dot{y} = 5x - y \quad (5.1.36)$$

1. Find the fixed point(s) of the system
2. Determine the stability of the fixed points(s)
3. Sketch the phase portrait.

Solution 5.1.5: Saddle

First we write it in matrix form, i.e.

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} -1 & 5 \\ 5 & -1 \end{pmatrix}. \quad (5.1.37)$$

There is only one fixed-point (check!) when $\dot{\mathbf{x}} = 0$ which is at the origin, i.e. $\mathbf{x}_e = \mathbf{0}$. The eigenvalues (check!) are

$$\sigma = -6, \quad 4, \quad (5.1.38)$$

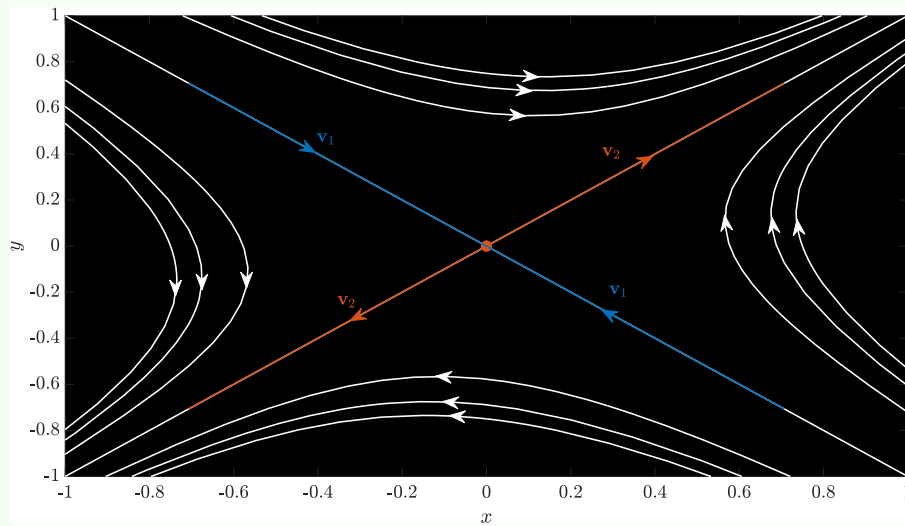
with eigenvectors

$$\mathbf{v}_1 = (1, -1)^T, \quad \mathbf{v}_2 = (1, 1)^T, \quad (5.1.39)$$

respectively, which means the solutions are of the form

$$\mathbf{x} = A\mathbf{v}_1 e^{-6t} + B\mathbf{v}_2 e^{4t} \quad (5.1.40)$$

with A, B arbitrary. In general, solutions will blow-up as $t \rightarrow \infty$ unless $B = 0$ and the initial conditions lie precisely on the eigenvalue $\mathbf{v}_1$ and then the solutions decay to zero. This is an example of a **saddle** node; trajectories blow-up except for two unique trajectories on the stable eigenmode, as shown in the figure below:

**Example 5.1.6: Centre**

Consider the dynamical system

$$\dot{x} = y \quad (5.1.41)$$

$$\dot{y} = -x \quad (5.1.42)$$

1. Find the fixed point(s) of the system
2. Determine the stability of the fixed point(s)
3. Sketch the phase portrait.

Solution 5.1.6: Centre

First we write it in matrix form, i.e.

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (5.1.43)$$

There is only one fixed-point (check!) when $\dot{\mathbf{x}} = 0$ which is at the origin, i.e. $\mathbf{x}_e = \mathbf{0}$.

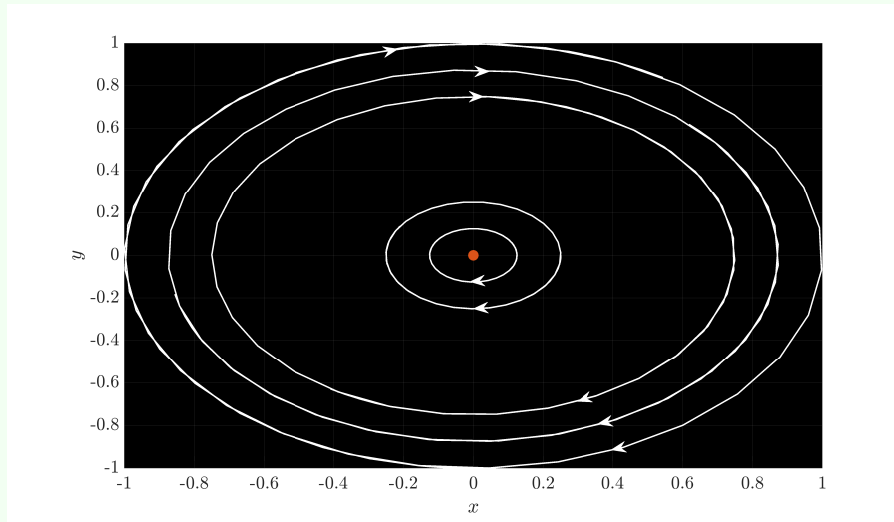
The eigenvalues (check!) are

$$\sigma = \pm i, \quad (5.1.44)$$

which are purely imaginary. Therefore the solutions are of the form

$$\mathbf{x} = A \cos t + B \sin t, \quad (5.1.45)$$

with A, B arbitrary. Solutions will oscillate and this is an example of a **center** node; trajectories form closed curves around the fixed-point as shown in the figure below



Example 5.1.7: Comb

Consider the dynamical system

$$\dot{x} = -3x - 6y \quad (5.1.46)$$

$$\dot{y} = -x - 2y \quad (5.1.47)$$

1. Find the fixed point(s) of the system
2. Determine the stability of the fixed points(s)
3. Sketch the phase portrait.

Solution 5.1.7: Comb

First we write it in matrix form, i.e.

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} -3 & -6 \\ -1 & -2 \end{pmatrix}. \quad (5.1.48)$$

There is only one fixed-point (check!) when $\dot{\mathbf{x}} = \mathbf{0}$ which is at the origin, i.e. $\mathbf{x}_e = \mathbf{0}$.

The eigenvalues (check!) are

$$\sigma = -5, \quad 0, \quad (5.1.49)$$

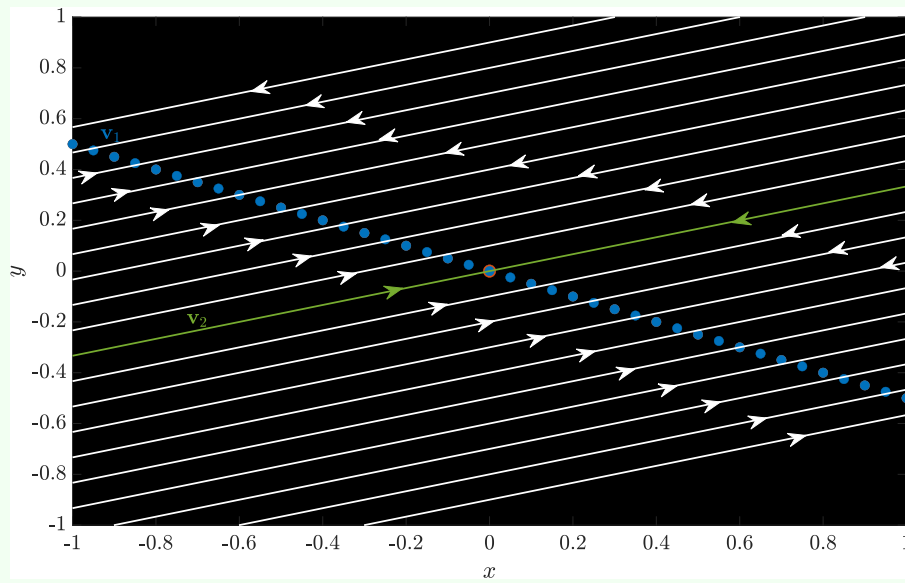
with eigenvectors

$$\mathbf{v}_1 = (1, 3)^T, \quad \mathbf{v}_2 = (1, -2)^T, \quad (5.1.50)$$

respectively. This has a solution of the form

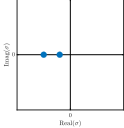
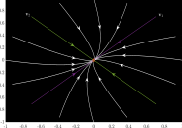
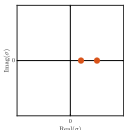
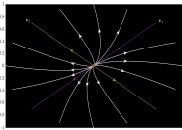
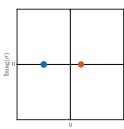
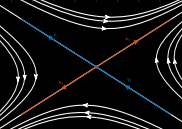
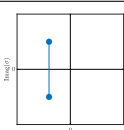
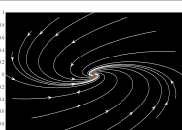
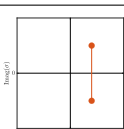
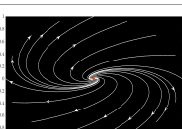
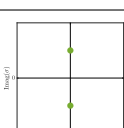
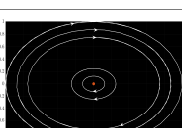
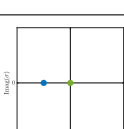
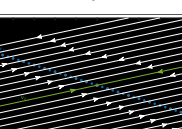
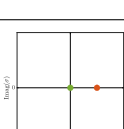
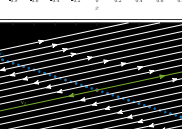
$$\mathbf{x} = A\mathbf{v}_1 + B\mathbf{v}_2 e^{-5t}, \quad (5.1.51)$$

for arbitrary A, B . This case is interesting because one of the eigenvalues is zero. This means there is a line of fixed-points in the direction $\mathbf{v}_1$ and all trajectories tend to this line in the direction of $\mathbf{v}_2$. This is an example of a **stable comb** and the phase portrait is shown below:



5.2. Linear two-dimensional dynamical systems: Summary

The stability of the fixed-point and the global phase portrait depends on the nature of the eigenvalues, σ , of the matrix $\mathbf{A}$ in (5.0.1) and is summarised in the table below. Key is the identification of the number of eigenvalues with real part positive, zero and negative, denoted by the vector (n^+, n^0, n^-) :

Type	(n^-, n^0, n^+)	Spectrum	Phase-portrait	Stability
Stable node	$(2, 0, 0)$			Asymptotically stable
Unstable node	$(0, 0, 2)$			Unstable
Saddle	$(1, 0, 1)$			Unstable
Stable spiral	$(2, 0, 0)$			Asymptotically stable
Unstable spiral	$(0, 0, 2)$			Unstable
Centre	$(2, 0, 0)$			Lyapunov stable
Stable comb	$(1, 1, 0)$			Lyapunov stable
Unstable comb	$(0, 1, 1)$			Unstable

1. If both of the eigenvalues are real and negative the phase space is a **stable node**.
2. If both of the eigenvalues are real and positive the phase space is a **unstable node**.
3. If both of the eigenvalues are complex with negative real part the phase space is a **stable spiral**.
4. If both of the eigenvalues are complex with positive real part the phase space is a **unstable spiral**.
5. If both of the eigenvalues are real but one is positive and the other is negative the phase space is a **saddle**.
6. If both of the eigenvalues are purely imaginary the phase space is a **center**.
7. If one of the eigenvalues is zero the phase space is a **stable/unstable comb**.